Block practice · Trace tables: running, reading and fixing algorithms

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Trace tables: run the algorithm by hand

Say what an algorithm (or one line) does

Find and fix errors in an algorithm

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Practice workbook · Trace tables: run the algorithm by hand

Executing a supplied flowchart, pseudocode algorithm or short program by hand for given inputs to find the outputs, the final or returned value of a variable, or to fill a trace table or output table.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-01

1. Question 4 · part c.i · 4 marks

2022 JUN · Paper 01

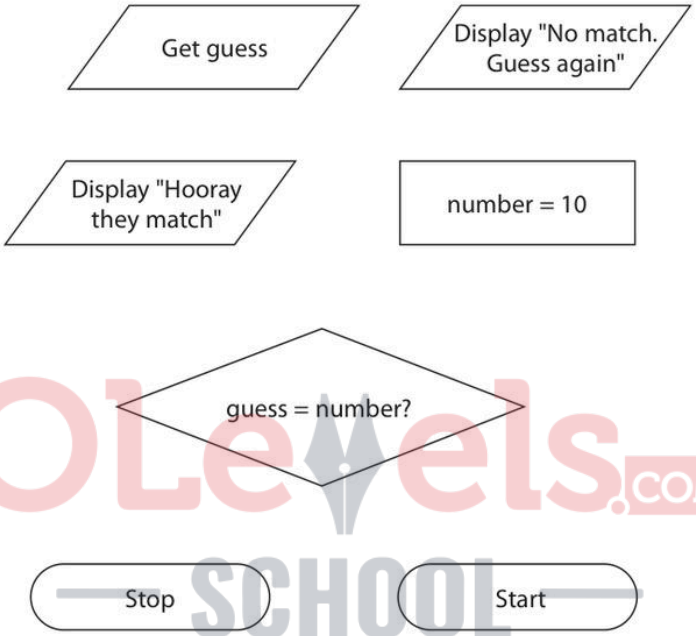
Worked-example retry · Attempt part c.i only.

4 Reba likes writing programs.

(a) She is writing a guessing game.

She needs a flowchart to show the logic of the game.

(i) These are the components needed to draw the flowchart.



Draw the flowchart for the algorithm in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

10

P 7 2 5 3 8 A 0 1 0 2 0

1. Question 4 · part c.i · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Draw your flowchart here.

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1. Question 4 · part c.i · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

(ii) Identify an alternative method for writing the algorithm.

(1)

- ☒ **A** Simulation
- ☒ **B** Cipher
- ☒ **C** Program code
- ☒ **D** Truth table

(b) Reba wants to develop a program that will convert a temperature in Fahrenheit to Celsius.

Here are four steps in the algorithm.

The steps are not in the correct order.

	Step
A	Change the temperature to Celsius
B	Get the temperature in Fahrenheit
C	Show the temperature in Celsius
D	Set the temperature to 0

(i) Give the letter of the step that initialises a variable.

(1)

(ii) Give the letter of the step that inputs a value.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 2 5 3 8 A 0 1 2 2 0

1. Question 4 · part c.i · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

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QUESTION 4 (c) BEGINS ON THE NEXT PAGE.

1. Question 4 · part c.i · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

(c) **Figure 2** shows the pseudocode for an early version of an algorithm that Reba has written for another game.

The algorithm:

- asks the user to input a colour or input –1 to end the game
- awards 1 point for red
- awards 8 points for orange
- generates the score for the game
- displays the results of the game.

```
SET Colour TO ""
SET Score TO 0
SET RedPoints TO 0
SET OrangePoints TO 0
SET NumOranges TO 0

WHILE Colour <> "-1" DO
   RECEIVE Colour FROM (STRING) KEYBOARD
   IF Colour = "red" THEN
      SET RedPoints TO RedPoints + 1
   ELSE
      IF Colour = "orange" THEN
         SET OrangePoints TO OrangePoints + 8
         SET NumOranges TO NumOranges + 1
      END IF
   END IF
END WHILE

SET Score TO RedPoints + OrangePoints

SEND ("Score: "& Score) TO DISPLAY
SEND ("Number of reds: "& RedPoints) TO DISPLAY
SEND ("Number of oranges: "& OrangePoints) TO DISPLAY
```

Figure 2

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 2 5 3 8 A 0 1 4 2 0

1. Question 4 · part c.i · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

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Reba inputs: red, orange, red, red, orange, –1

The outputs are not as she expects.

(i) Complete the trace table to show the outputs.

(4)

Colour	Score	RedPoints	OrangePoints	NumOranges	Outputs
	0	0	0	0	
red					
orange					
red					
red					
orange					
–1					

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(ii) Give the line number of the pseudocode that contains the error.

(1)

.....

(iii) Write a replacement line of pseudocode to correct the error.

(1)

.....

(Total for Question 4 = 14 marks)

1. Question 4 · part c.i · 4 marks · Working and answer

Working and answer

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2. Question 5 · part a.i · 3 marks

2019 JUN · Paper 01

Further practice · Attempt part a.i only.

5 Ships carry cargo around the world in containers.

(a) Containers come in two sizes.

Figure 2 shows an algorithm written using flowchart symbols.

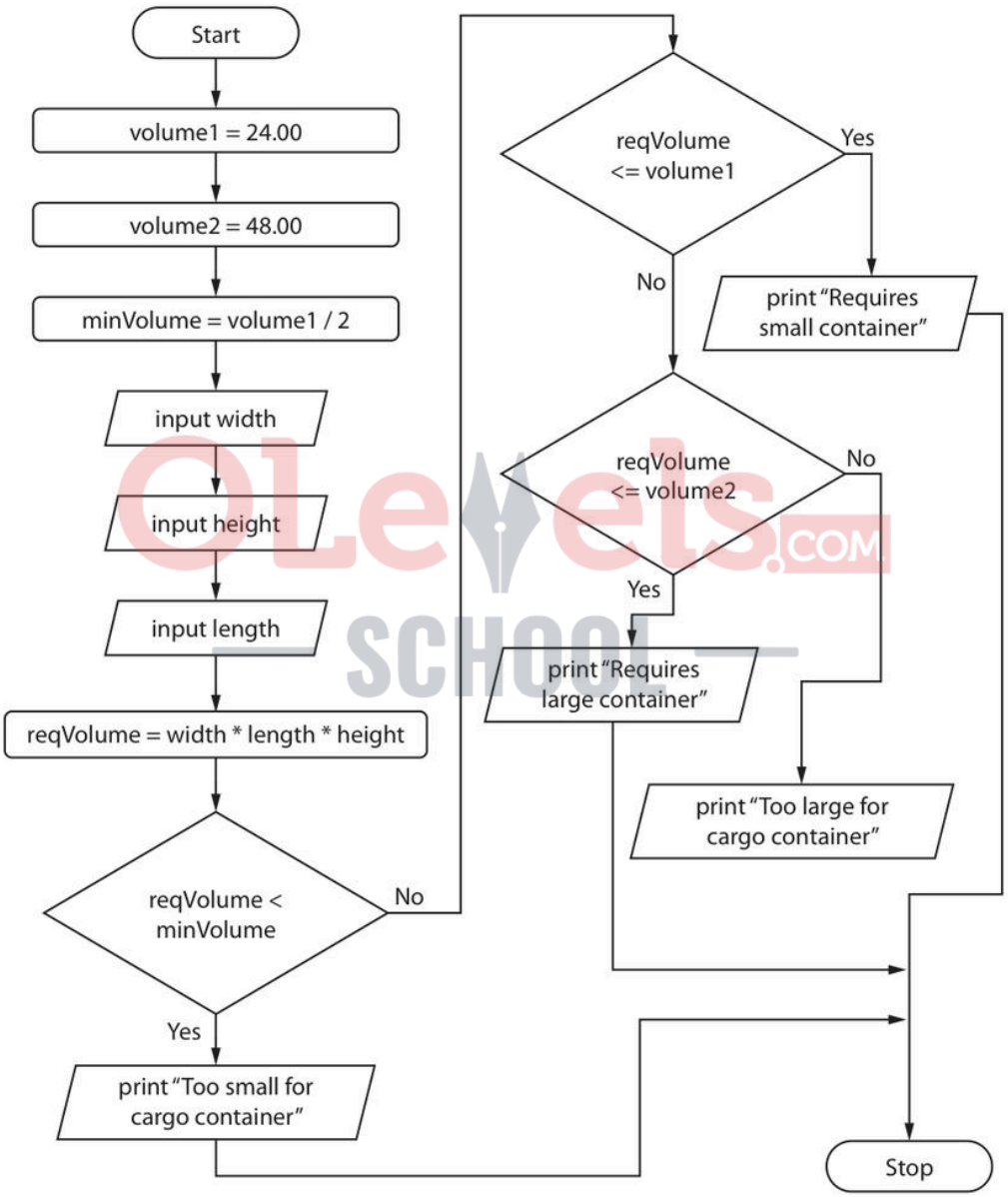


Figure 2

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DO NOT WRITE IN THIS AREA

2. Question 5 · part a.i · 3 marks

2019 JUN · Paper 01

Further practice · Attempt part a.i only.

(i) Complete the table to show the output for each cargo item.

(3)

Cargo item			Output
width	length	height	
4	4	2	
2	2	2	
3	8	5	

(ii) State the purpose of the algorithm in **Figure 2**.

(1)

DO NOT WRITE IN THIS AREA

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2. Question 5 · part a.i · 3 marks · Working and answer

Working and answer

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3. Question 5 · part c.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.i only.

5 Ships carry cargo around the world in containers.

(a) Containers come in two sizes.

Figure 2 shows an algorithm written using flowchart symbols.

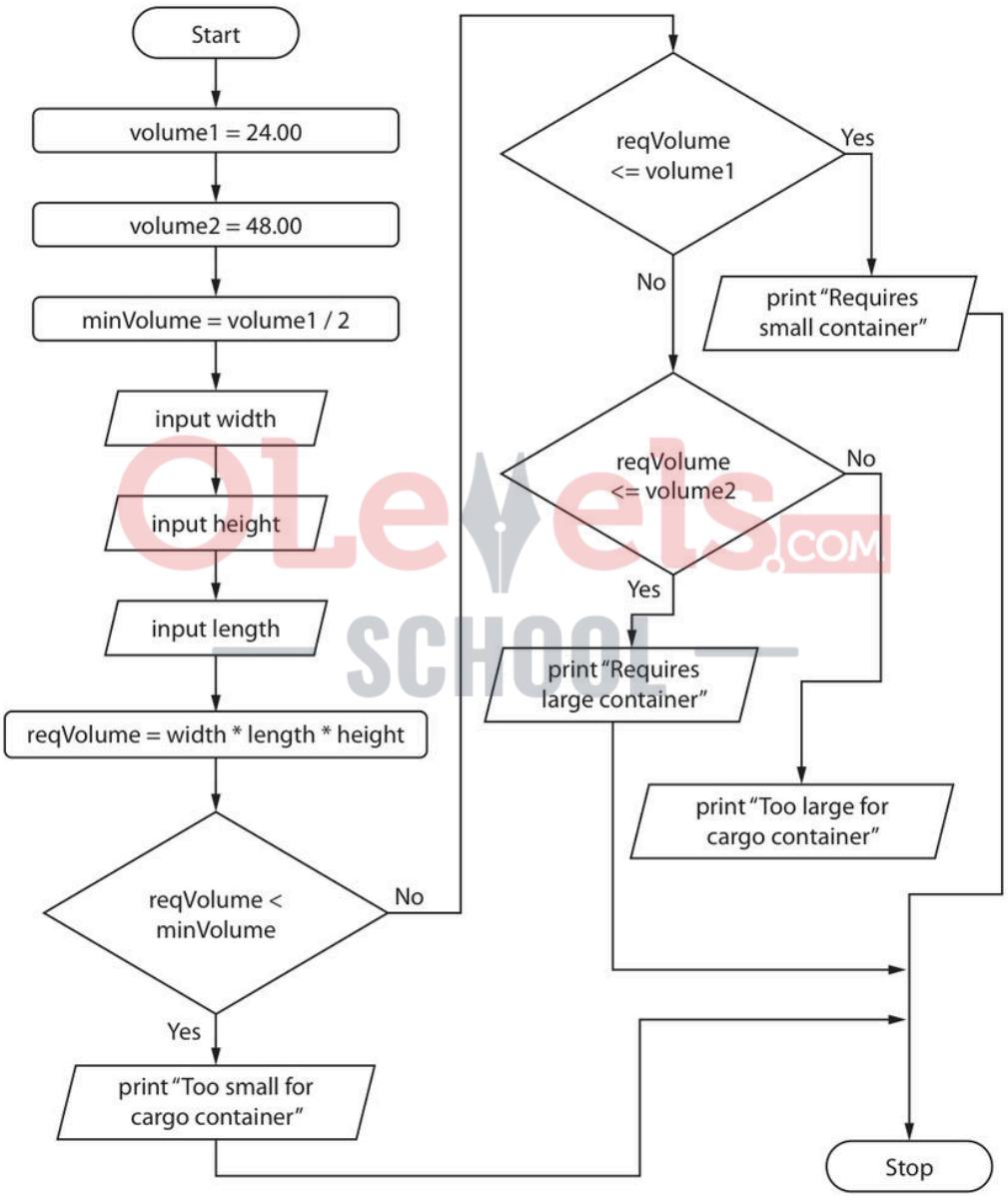


Figure 2

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3. Question 5 · part c.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.i only.

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(i) Complete the table to show the output for each cargo item.

(3)

Cargo item			Output
width	length	height	
4	4	2	
2	2	2	
3	8	5	

(ii) State the purpose of the algorithm in **Figure 2**.

(1)

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3. Question 5 · part c.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.i only.

(b) Each ship is registered to a state.

Figure 3 shows a partially completed algorithm written in pseudocode.

The completed algorithm must:

- print each state to the display on a new line
- count the number of states
- create a message as a single string (e.g. there are *number* states)
- print the message to the display.

Complete the algorithm in the space provided in **Figure 3**.

(4)

```
SET numStates TO 0
SET states TO ["France", "Singapore", "Malta", "Panama", "Greece", "Italy"]

FOR EACH state FROM states DO




END FOREACH
```

Figure 3

DO NOT WRITE IN THIS AREA
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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

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3. Question 5 · part c.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.i only.

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3. Question 5 · part c.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.i only.

(c) Cargo ships have maximum weight loads.

Figure 4 shows an algorithm written using pseudocode.

The algorithm should identify the size of cargo ship required for any load.

There is an error on line 9.

```
2   SET loadWeight TO [20000, 28000, 40000, 50000]
3   SET index TO 0
4   SET found TO FALSE
5
6   SEND "Enter cargo weight" TO DISPLAY
7   RECEIVE target FROM (INTEGER) KEYBOARD
8
9   WHILE (NOT found) DO
10      IF (loadWeight [index] >= target) THEN
11          SEND loadWeight [index] TO DISPLAY
12          SET found TO TRUE
13      ELSE
14          SET index TO  index + 1
15      END IF
16  END WHILE
17
18  IF (NOT found) THEN
19      SEND "No ship available" TO DISPLAY
20  END IF
```

Figure 4

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 1 8 8 0 A 0 1 8 2 4

3. Question 5 · part c.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.i only.

(i) Trace tables are used to identify errors in algorithms.

Complete the trace table for an input of 50500 to show what happens due to the error on line 9 in the pseudocode in **Figure 4**.

You may not need to fill in all the rows in the table.

(2)

target	found	index	loadWeight[index]
50500	FALSE	0	

(ii) Construct a single line of pseudocode to correct line 9.

(2)

.....

.....

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.....

(Total for Question 5 = 12 marks)

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3. Question 5 · part c.i · 2 marks · Working and answer

Working and answer

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4. Question 2 · part b.i · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.i only.

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2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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4. Question 2 · part b.i · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.i only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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4. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

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5. Question 2 · part b.iv · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.iv only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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5. Question 2 · part b.iv · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.iv only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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5. Question 2 · part b.iv · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.iv only.

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(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

(2)

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5. Question 2 · part b.iv · 1 mark · Working and answer

Working and answer

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6. Question 2 · part c.v · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.v only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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6. Question 2 · part c.v · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.v only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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6. Question 2 · part c.v · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.v only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

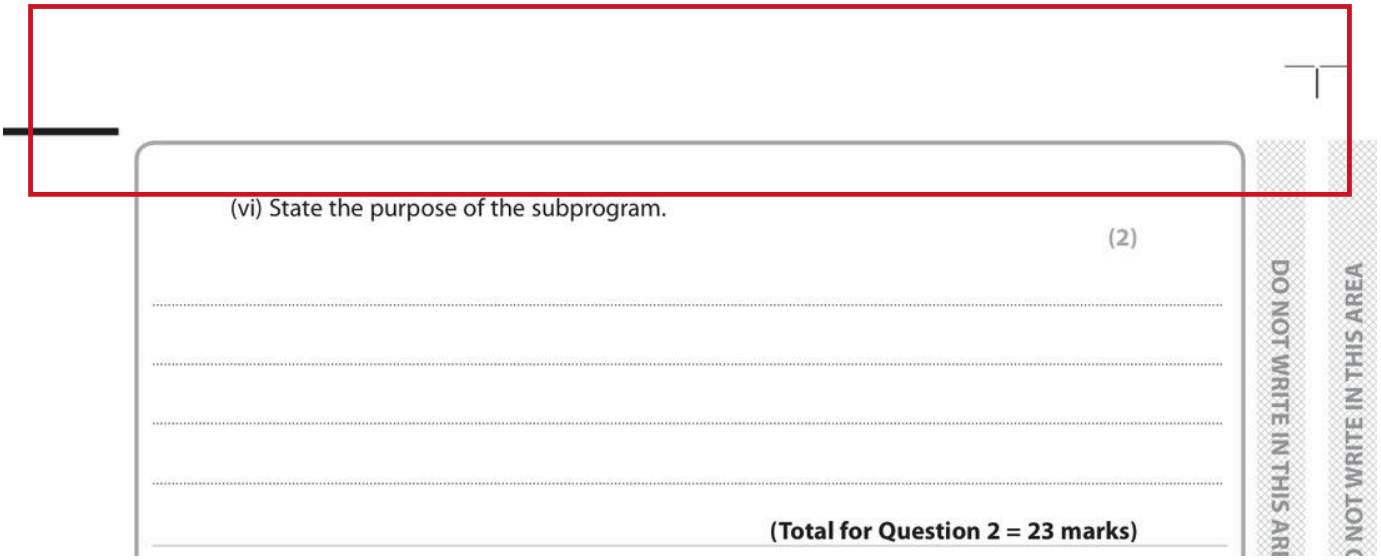
(2)

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6. Question 2 · part c.v · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.v only.


(vi) State the purpose of the subprogram.

(2)

(Total for Question 2 = 23 marks)

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Working and answer

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7. Question 6 · part b.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b.i only.

6 A software engineer is working with some algorithms.

(a) An algorithm needs to be developed.

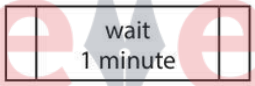
The algorithm must:

- check for a change of hour every minute
- use the library subprogram getHour() to get the hour part of the current time in the 24-hour clock (0 to 23)
- output "Good morning" when the hour is between 3 and 12, inclusive
- output "Good afternoon" when the hour is between 13 and 19, inclusive
- output "Good night" at all other times.

Complete the flowchart to represent this algorithm, in the space provided on the next page.

The call to the library subprogram getHour() has already been included.

Use this subprogram symbol to show waiting for 1 minute.



(6)

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DO NOT WRITE IN THIS AREA

7. Question 6 · part b.i · 2 marks

2020 NOV · Paper 01

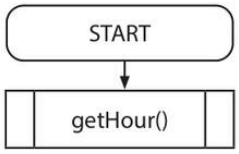
Further practice · Attempt part b.i only.

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Draw your flowchart here.



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7. Question 6 · part b.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b.i only.

(b) Another algorithm determines when to change the flavoured syrups for a drinks dispensing machine.

Figure 4 shows this algorithm written in pseudocode.

```
1  SET flavours TO ["anise", "mango", "cola", "apple", "papaya", "strawberry",
   "lychee", "banana"]
2
3  SET volume TO [0.7, 0.2, 0.6, 0.1, 0.05, 0.8, 0.4, 0.6]
4
5  SET i TO 0
6
7  WHILE i < 8 DO
8
9      IF (volume[i] < 0.1) THEN
10         SEND (flavours[i] & " needs changing") TO DISPLAY
11
12     ELSE
13         IF (volume[i] >= 0.3) AND (volume[i] <= 0.5) THEN
14             SEND (flavours[i] & " needs ordering")  TO DISPLAY
15         END IF
16     END IF
17
18     SET i TO i + 1
19
20 END WHILE
```

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Figure 4

(i) Give the output produced by the algorithm.

(2)

.....

.....

.....

.....

The algorithm works with a fixed number of flavours.

(ii) Give the number of the line in the algorithm that would need to be amended to allow for any number of flavours.

(1)

.....

.....

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7. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

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8. Question 5 · part a.i · 5 marks

2021 JUN · Paper 01

Further practice · Attempt part a.i only.

5 Isaac is a program developer.

(a) **Figure 3** shows an algorithm Isaac has written.

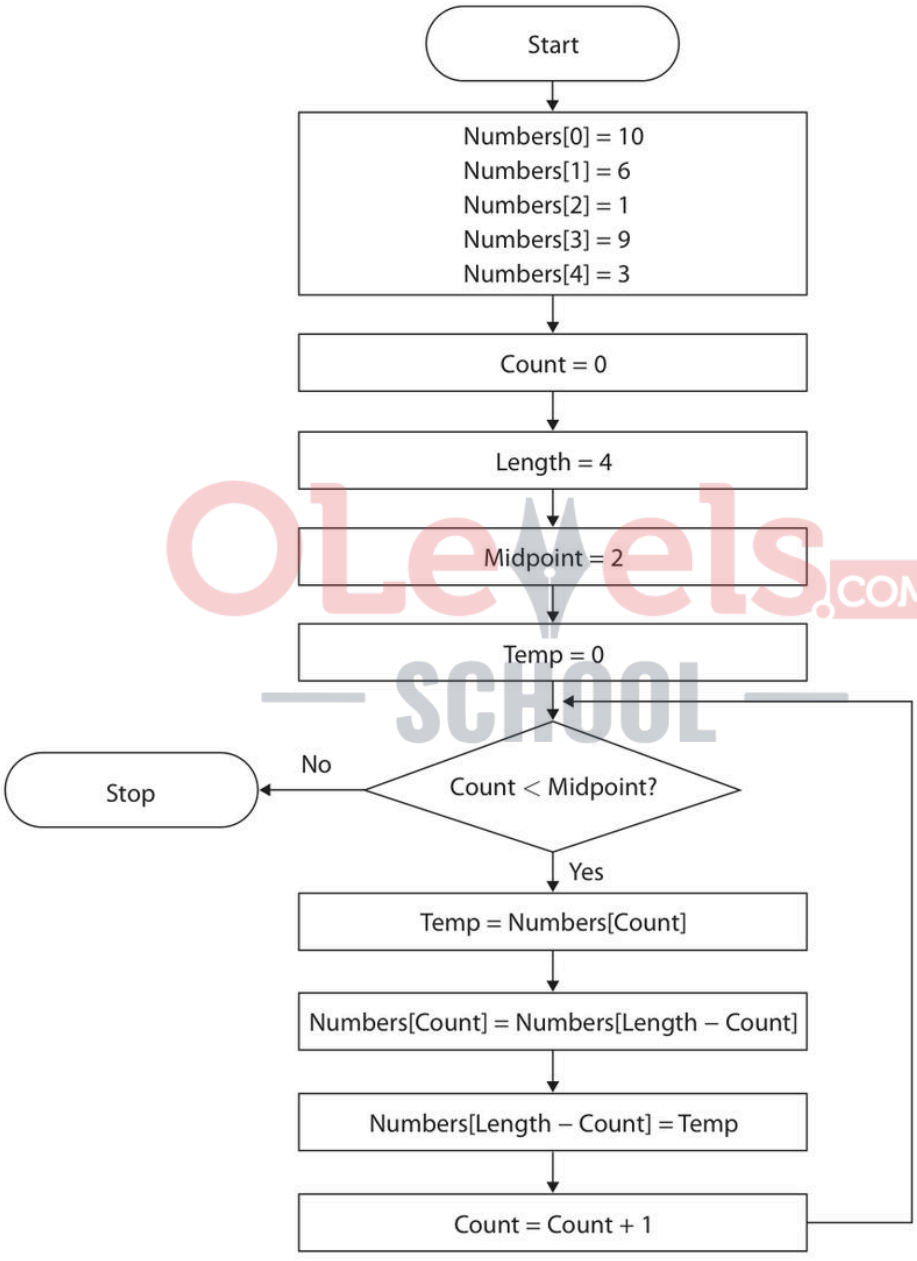


Figure 3

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8. Question 5 · part a.i · 5 marks

2021 JUN · Paper 01

Further practice · Attempt part a.i only.

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DO NOT WRITE IN THIS AREA

(i) Complete the trace table. You may not need to use all of the rows.

(5)

				Numbers array				
Count	**Length**	**Midpoint**	**Temp**	[0]	[1]	[2]	[3]	[4]
0	4	2	0	10	6	1	9	3

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(ii) Give the purpose of the algorithm.

(1)

(iii) Explain why the variable Temp is needed.

(2)

8. Question 5 · part a.i · 5 marks · Working and answer

Working and answer

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9. Question 3 · part d.i · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.i only.

3 Several encryption algorithms have been developed.

(a) Ahmed has started to create a Caesar cipher encryption program.

Caesar cipher encryption works by giving a number value to a key.

Each letter in a plaintext message is replaced by a new ciphertext letter using the key.

For example, as shown in the table, a key value of +2 would change the plaintext message **encrypt** to the ciphertext message **gpetarv**.

Plaintext	e	n	c	r	y	p	t
Ciphertext	g	p	e	t	a	r	v

Ahmed's program already converts from plaintext to ciphertext.

He wants to extend the program to:

- accept the input of a plaintext message
- accept the input of the key, which must be a positive integer between 1 and 25
- write the generated ciphertext to a text file named **Cipher.txt**
- display the ciphertext.

Open **Q03a** in the code editor.

Complete the program code.

You must use the structure given in the file **Q03a** to complete the program.

Do not add any further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(8)

(b) Explain **one** reason why the Caesar cipher is a weak encryption algorithm.

(2)

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9. Question 3 · part d.i · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.i only.

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(c) The Vigenere cipher encrypts plaintext using a series of Caesar ciphers, based on the letters of a keyword and a Vigenere square.

Figure 1 shows a Vigenere square.

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
b	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a
c	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b
d	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c
e	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d
f	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e
g	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f
h	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g
i	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h
j	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i
k	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j
l	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k
m	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l
n	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m
o	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n
p	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
q	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p
r	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q
s	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r
t	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s
u	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t
v	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u
w	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v
x	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
y	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x
z	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y

Figure 1

Complete the table to perform a Vigenere decryption of the ciphertext message **flmktrwhee** using the keyword **thirty**.

(4)

Encrypted letter	f	l	m	k	t	r	w	h	e	e
Keyword letter										
Decrypted letter										

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9. Question 3 · part d.i · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.i only.

(d) A user-defined function uses a Vigenere cipher to encrypt plaintext into ciphertext.

This pseudocode shows how the function is called.

```
SET ciphertext TO ciphertext & getCipherLetter(keywordLetter, plaintextLetter)
```

(i) State what would be returned by this function.

(1)

(ii) Give the name of **one** parameter passed to this function.

(1)

(iii) State what is meant by a **built-in subprogram**.

(1)

(Total for Question 3 = 17 marks)

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Working and answer

10. Question 4 · part b.i · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part b.i only.

4 Algorithms can be used to perform calculations and to process data.

(a) State what is meant by the term **algorithm**.

(1)

.....

.....

(b) **Figure 3** shows an algorithm.

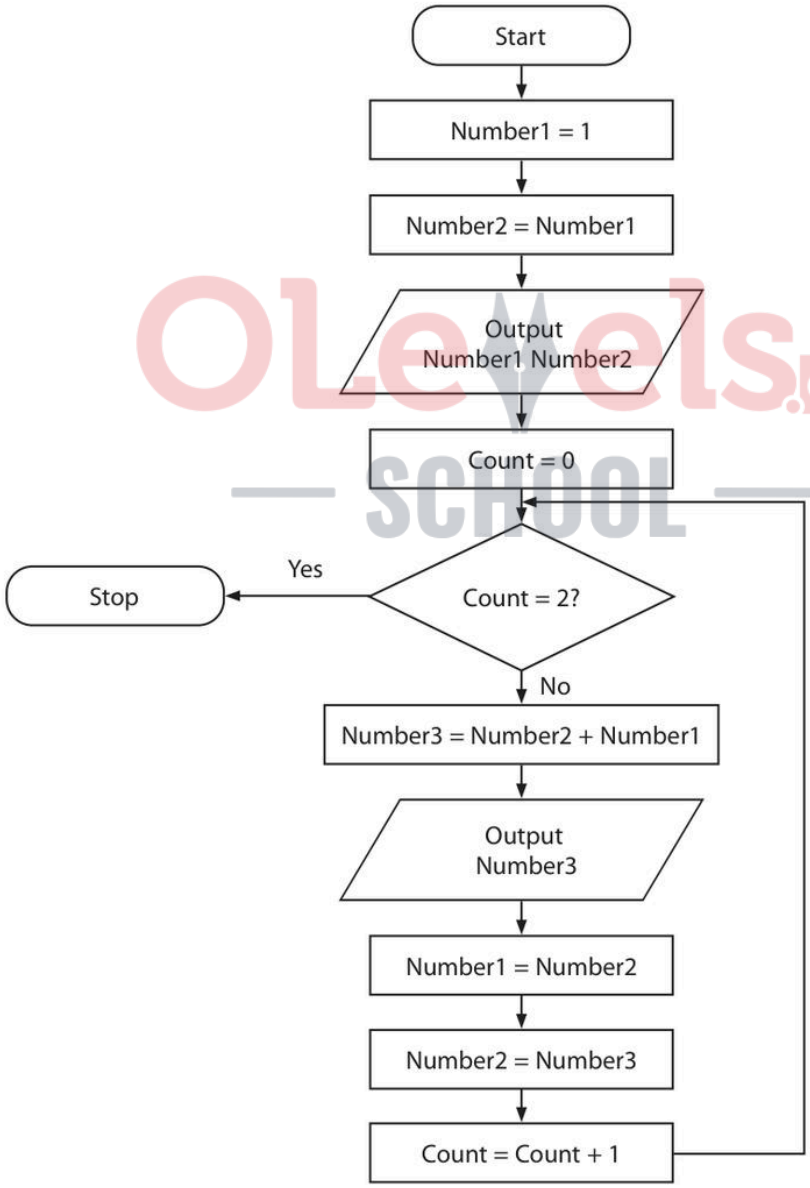


Figure 3

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P 6 9 2 9 7 A 0 1 2 2 0

10. Question 4 · part b.i · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part b.i only.

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(i) Complete this trace table for the algorithm.

You may not need to use all the rows.

(5)

Number1	Number2	Number3	Count	Output	Count = 2?
1	1	–	0	1 1	False

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(ii) The benefits of using a trace table include that they allow variable states, outputs and decisions to be recorded.

Give **one other** benefit of using a trace table to test an algorithm.

(1)

.....

.....

10. Question 4 · part b.i · 5 marks · Working and answer

Working and answer

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11. Question 3 · part b · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

3 Algorithms can be represented in flowcharts, pseudocode or program code.

(a) Trace tables can be used with flowcharts or pseudocode.

Give **two** characteristics of a trace table.

(2)

1 .....

.....

2 .....

.....

(b) An algorithm has been written to validate numbers entered by the user.

Figure 2 shows the pseudocode for the algorithm.

```
1  RECEIVE num1 FROM (INTEGER) KEYBOARD
2  RECEIVE num2 FROM (INTEGER) KEYBOARD
3  IF (((num1 < 16) AND (num2 < 23)) OR (num2 = 13)) THEN
4      SEND "State 1" TO DISPLAY
5  ELSE
6      IF ((num2 > 12) AND (num1 > 20)) THEN
7          SEND "State 2" TO DISPLAY
8      ELSE
9          IF ((num1 = 88) OR NOT (num2 = 18)) THEN
10             SEND "State 3" TO DISPLAY
11         ELSE
12             SEND "State 4" TO DISPLAY
13         END IF
14     END IF
15 END IF
```

Figure 2

Complete the table to show the output for each set of inputs.

(3)

num1	num2	Output
88	18	
17	18	
12	19	

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P 7 2 9 3 8 A 0 6 1 6

11. Question 3 · part b · 3 marks

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Further practice · Attempt part b only.

(c) A program is required to calculate the result of raising one integer (the base) to the power of another (the exponent).

Figure 3 shows a flowchart for the algorithm.

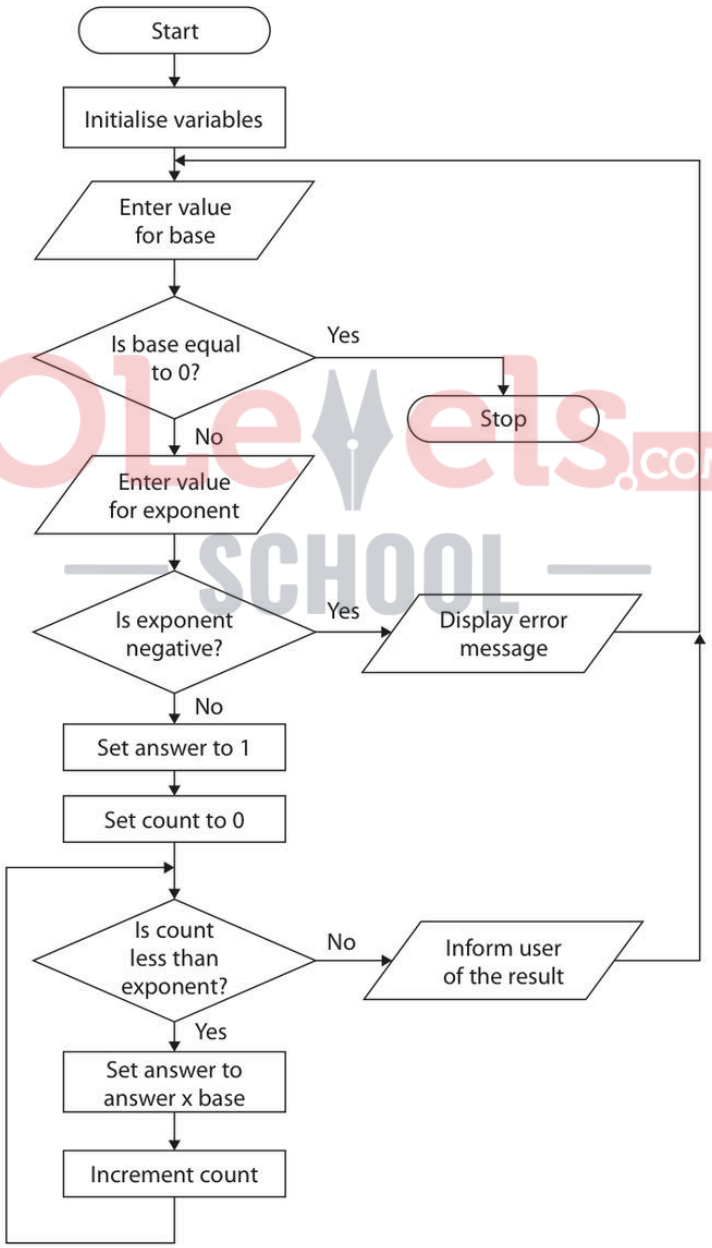


Figure 3

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11. Question 3 · part b · 3 marks · Working and answer

Working and answer

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12. Question 6 · part b.i · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

6 Algorithms can be represented using flowcharts or pseudocode.

(a) An algorithm has been designed to perform these steps:

- A user is asked to input a number **n**
- If the number **n** is 0 the program stops
- If the number **n** is positive a message to display 'Positive' is output
- If the number **n** is negative a message to display 'Negative' is output
- The program repeats until a value of 0 has been input.

Draw a flowchart to represent this algorithm.

(6)

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12. Question 6 · part b.i · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

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(b) **Figure 3** shows an algorithm written in pseudocode that performs a mathematical process.

```
01
02  RECEIVE number FROM (INTEGER) KEYBOARD
03
04  SET flag TO FALSE
05
06  IF number = 1 THEN
07     SET flag TO TRUE
08  ELSE
09     SET testNum TO 1
10     REPEAT
11        SET testNum TO testNum + 1
12        IF (number MOD testNum) = 0 THEN
13           SET flag TO TRUE
14        END IF
15     UNTIL (testNum = number) OR (flag = TRUE)
16  END IF
17
18  IF NOT (flag = TRUE)
19     SEND number TO DISPLAY
20  END IF
21
```

Figure 3

Trace tables are used to show the outputs from algorithms.

(i) Complete the trace table for an input of 7

(2)

number	flag	testNum	DISPLAY

12. Question 6 · part b.i · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

(ii) Complete the trace table for an input of 9

(2)

number	flag	testNum	DISPLAY

(iii) State the purpose of the algorithm.

(1)

(iv) Lines 10 to 15 of the pseudocode in **Figure 3** use a post-conditioned loop.

Explain **one** reason why it is better for the algorithm to use a post-conditioned loop rather than a count controlled loop.

(2)

(Total for Question 6 = 13 marks)

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12. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

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13. Question 6 · part b.ii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.ii only.

6 Algorithms can be represented using flowcharts or pseudocode.

(a) An algorithm has been designed to perform these steps:

- A user is asked to input a number **n**
- If the number **n** is 0 the program stops
- If the number **n** is positive a message to display 'Positive' is output
- If the number **n** is negative a message to display 'Negative' is output
- The program repeats until a value of 0 has been input.

Draw a flowchart to represent this algorithm.

(6)

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13. Question 6 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

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(b) **Figure 3** shows an algorithm written in pseudocode that performs a mathematical process.

```
01
02  RECEIVE number FROM (INTEGER) KEYBOARD
03
04  SET flag TO FALSE
05
06  IF number = 1 THEN
07     SET flag TO TRUE
08  ELSE
09     SET testNum TO 1
10     REPEAT
11        SET testNum TO testNum + 1
12        IF (number MOD testNum) = 0 THEN
13           SET flag TO TRUE
14        END IF
15     UNTIL (testNum = number) OR (flag = TRUE)
16  END IF
17
18  IF NOT (flag = TRUE)
19     SEND number TO DISPLAY
20  END IF
21
```

Figure 3

Trace tables are used to show the outputs from algorithms.

(i) Complete the trace table for an input of 7

(2)

number	flag	testNum	DISPLAY

13. Question 6 · part b.ii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.ii only.

(ii) Complete the trace table for an input of 9

(2)

number	flag	testNum	DISPLAY

(iii) State the purpose of the algorithm.

(1)

(iv) Lines 10 to 15 of the pseudocode in **Figure 3** use a post-conditioned loop.

Explain **one** reason why it is better for the algorithm to use a post-conditioned loop rather than a count controlled loop.

(2)

(Total for Question 6 = 13 marks)

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13. Question 6 · part b.ii · 2 marks · Working and answer

Working and answer

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Practice workbook · Say what an algorithm (or one line) does

Reading a supplied flowchart, pseudocode algorithm, subprogram or opened program and explaining in words either its overall job or the role of a particular line or variable (a temporary swap variable, an index increment, a redundant test, a count-down loop variable).

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-02

1. Question 5 · part a.ii · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part a.ii only.

5 Martha is a computer science student and is learning to code.

(a) **Figure 7** shows an algorithm in pseudocode written by Martha.

```
1  SET first TO 0
2  SET second TO 0
3  SET answer TO 0
4  SET user_answer TO 0
5  SET score TO 0
6
7  REPEAT 10 TIMES
8      SET score TO 0
9      SET first TO RANDOM(99)
10     SET second TO RANDOM(99)
11     SET answer TO first + second
12
13     SEND first "+" second "=" TO DISPLAY
14     RECEIVE user_answer FROM (INTEGER) KEYBOARD
15
16     IF user_answer = answer THEN
17        SET score TO score + 1
18     END IF
19 END REPEAT
20 SEND score TO DISPLAY
```

Figure 7

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
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1. Question 5 · part a.ii · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part a.ii only.

There is a logic error on line 8 of the pseudocode.

(i) Explain the problem this error will cause.

(2)

(ii) Describe the purpose of the algorithm shown in **Figure 7**, assuming the error has been corrected.

(2)

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1. Question 5 · part a.ii · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part a.ii only.

(b) Martha is writing a program to calculate grades.

Each student takes three tests. Each test has a maximum score of 50.

Scores are stored in a two-dimensional array called tbl_scores.

The lowest of the three scores is ignored and the other two are added to get a total out of 100.

Grades are based on the table in **Figure 8**.

Total	Grade
80 or more	Distinction
65 to 79	Merit
50 to 64	Pass
49 or less	Fail

Figure 8

Each student's total and grade are saved to a text file.

For example, the first student scores 45, 42 and 43. The lowest score 42 is ignored. The total is 88 and the grade is Distinction.

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1. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

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2. Question 5 · part a.ii · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part a.ii only.

5 Ships carry cargo around the world in containers.

(a) Containers come in two sizes.

Figure 2 shows an algorithm written using flowchart symbols.

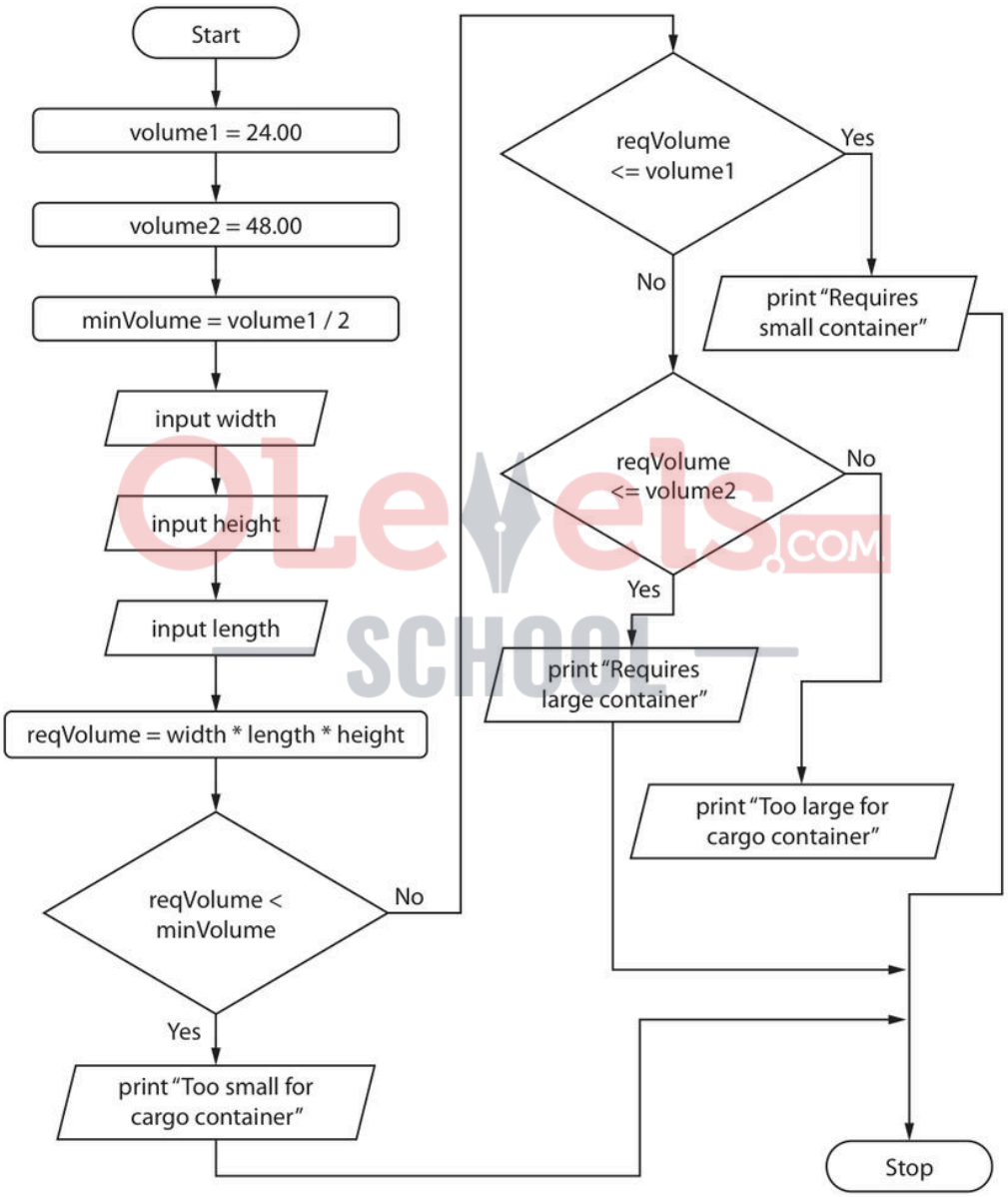


Figure 2

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P 6 1 8 8 0 A 0 1 4 2 4

2. Question 5 · part a.ii · 1 mark

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Further practice · Attempt part a.ii only.

(i) Complete the table to show the output for each cargo item.

(3)

Cargo item			Output
width	length	height	
4	4	2	
2	2	2	
3	8	5	

(ii) State the purpose of the algorithm in **Figure 2**.

(1)

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2. Question 5 · part a.ii · 1 mark

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Further practice · Attempt part a.ii only.

(b) Each ship is registered to a state.

Figure 3 shows a partially completed algorithm written in pseudocode.

The completed algorithm must:

- print each state to the display on a new line
- count the number of states
- create a message as a single string (e.g. there are *number* states)
- print the message to the display.

Complete the algorithm in the space provided in **Figure 3**.

(4)

```
SET numStates TO 0
SET states TO ["France", "Singapore", "Malta", "Panama", "Greece", "Italy"]

FOR EACH state FROM states DO



END FOREACH
```

Figure 3

DO NOT WRITE IN THIS AREA
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DO NOT WRITE IN THIS AREA

P 6 1 8 8 0 A 0 1 6 2 4

2. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

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3. Question 2 · part c.vi · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.vi only.

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2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

3. Question 2 · part c.vi · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.vi only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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3. Question 2 · part c.vi · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.vi only.

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(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

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(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

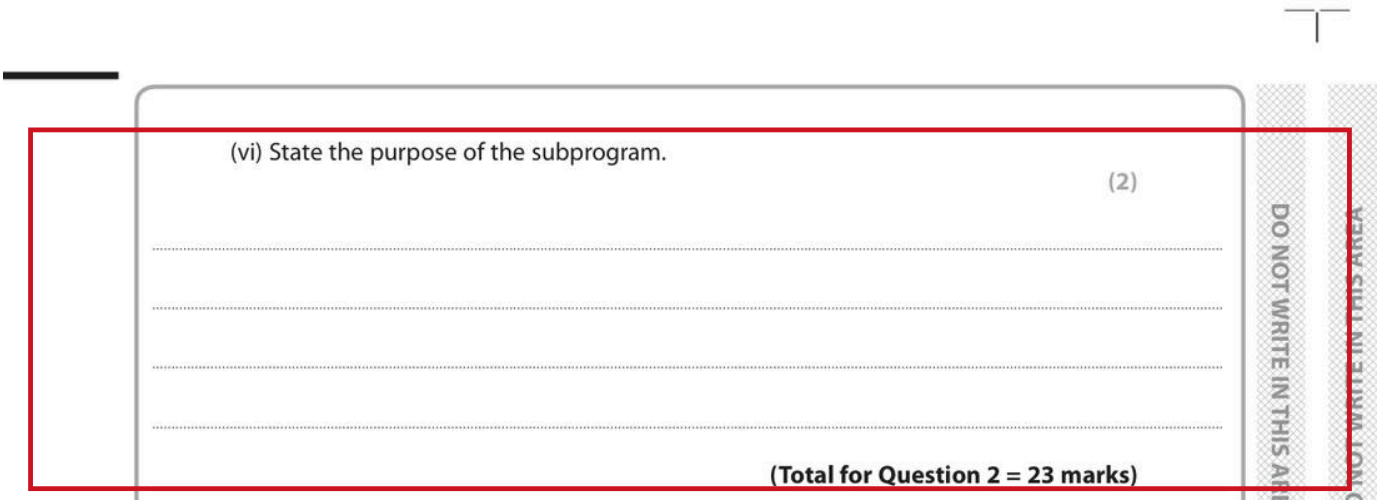
(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

(2)

3. Question 2 · part c.vi · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.vi only.



(vi) State the purpose of the subprogram.

(2)

(Total for Question 2 = 23 marks)

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Working and answer

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4. Question 5 · part a.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part a.ii only.

5 Isaac is a program developer.

(a) **Figure 3** shows an algorithm Isaac has written.

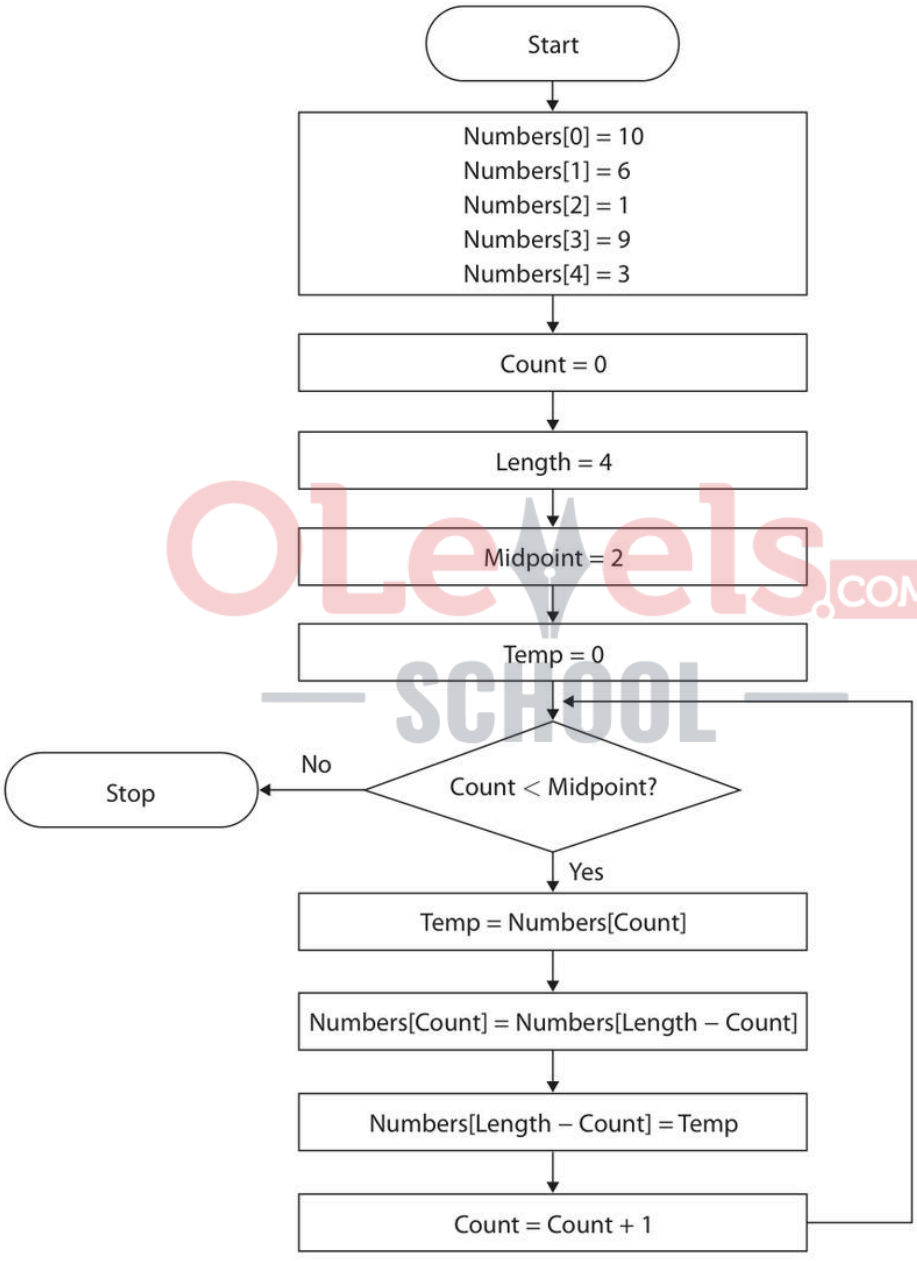


Figure 3

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P 6 6 4 8 9 A 0 1 2 2 0

4. Question 5 · part a.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part a.ii only.

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(i) Complete the trace table. You may not need to use all of the rows.

(5)

				Numbers array				
Count	**Length**	**Midpoint**	**Temp**	[0]	[1]	[2]	[3]	[4]
0	4	2	0	10	6	1	9	3

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(ii) Give the purpose of the algorithm.

(1)

.....

.....

(iii) Explain why the variable Temp is needed.

(2)

.....

.....

.....

.....

4. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

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5. Question 5 · part a.iii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part a.iii only.

5 Isaac is a program developer.

(a) **Figure 3** shows an algorithm Isaac has written.

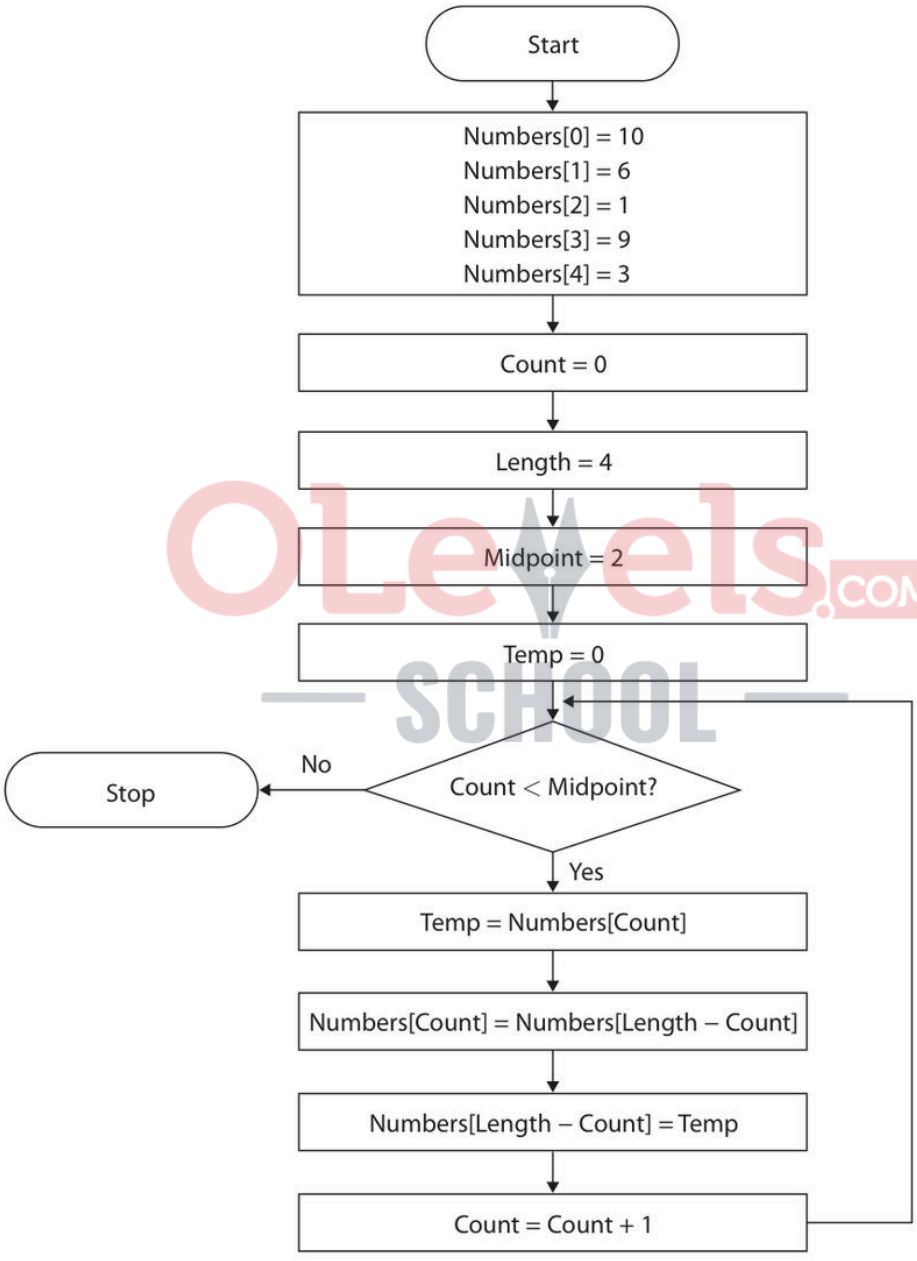


Figure 3

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P 6 6 4 8 9 A 0 1 2 2 0

5. Question 5 · part a.iii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(i) Complete the trace table. You may not need to use all of the rows.

(5)

				Numbers array				
Count	**Length**	**Midpoint**	**Temp**	[0]	[1]	[2]	[3]	[4]
0	4	2	0	10	6	1	9	3

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(ii) Give the purpose of the algorithm.

(1)

(iii) Explain why the variable Temp is needed.

(2)

5. Question 5 · part a.iii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part a.iii only.

(b) **Figure 4** shows an algorithm Isaac has written using pseudocode.

The algorithm should display the average of the numbers that have been input.

```
1   SET total TO 0
2   SET number TO 0
3   SET count TO 0
4   WHILE number <> –1 DO
5      SEND 'Input a number or –1 to end the program' TO DISPLAY
6      RECEIVE number FROM (INTEGER) KEYBOARD
7      SET total TO total + number
8      SET count TO count + 1
9   END WHILE
10  SET average TO total / count
11  SEND 'The average is ' & average TO DISPLAY
```

Figure 4

Isaac uses the input 2, 3, 5, 2, –1 to test the algorithm. He discovers an error.

Expected result	Actual result
The average is 2.75	The average is 2.2

(i) Explain why the **Actual** result is not the same as the **Expected** result.

(2)

(ii) Give the number of the line that contains the error.

(1)

(iii) Amend a single line of pseudocode to correct the error.

(1)

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14

P 6 6 4 8 9 A 0 1 4 2 0

5. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

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6. Question 4 · part b · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

4 Programs can be used to manipulate data.

(a) A program is needed to convert an 8-bit binary number into denary.

The program must:

- receive the input of a binary pattern and validate that it has exactly eight characters (there is no need to validate that it only contains 0s and 1s)
- request the input again if the binary pattern does not have exactly eight characters
- when a valid binary pattern is input:
  - convert the binary pattern into its denary equivalent
  - print a message that contains the binary pattern and its denary equivalent
- include a comment to explain how the conversion process works.

HINT

Denary placeholder	128	64	32	16	8	4	2	1
Binary number	1	0	1	1	0	1	0	0
Denary number	128	0	32	16	0	4	0	0

128 + 32 + 16 + 4 = 180

Open **Q04a** in the code editor.

Write a program to implement these requirements.

You must use the structure given in **Q04a** to write the program.

Do not add any further functionality.

Save your code as **Q04aFINISHED** with the correct file extension for the programming language.

(11)

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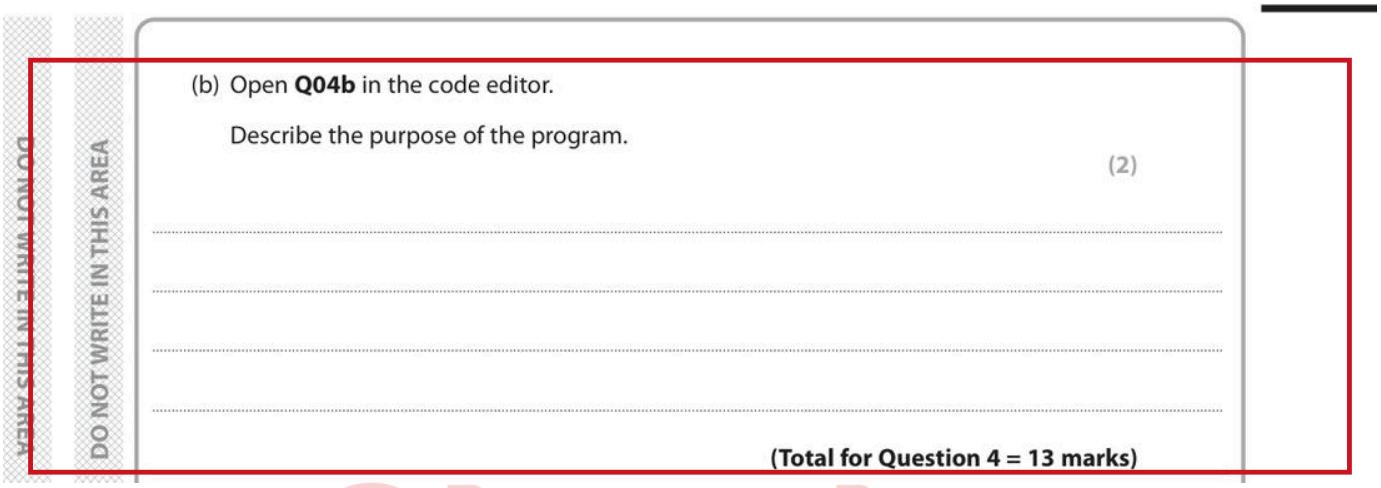
10

P 6 9 2 9 8 A 0 1 0 1 6

6. Question 4 · part b · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.



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(b) Open **Q04b** in the code editor.

Describe the purpose of the program.

(2)

(Total for Question 4 = 13 marks)

Working and answer

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7. Question 6 · part b · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b only.

6 Programmers share algorithms with different people and write algorithms for different reasons.

(a) A programmer is showing a new algorithm to a group of non-technical managers.

State an appropriate method for writing the algorithm.

Justify your answer.

(2)

Method

.....

Justification

.....

.....

.....

(b) **Figure 1** shows an algorithm that displays a string based on the number input by the user.

```
1  SEND ("Enter a number: ") TO DISPLAY
2  RECEIVE inNum FROM (INTEGER) KEYBOARD
3  IF ((inNum = 1) OR (inNum = 2)) THEN
4      IF (inNum = 1) THEN
5          SEND ("First") TO DISPLAY
6      ELSE
7          IF (inNum = 2) THEN
8              SEND ("Second") TO DISPLAY
9          END IF
10     END IF
11 ELSE
12     SEND ("Invalid input") TO DISPLAY
13 END IF
```

Figure 1

Give **one** reason why the selection statement on line 7 is not required.

(1)

.....

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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20

P 7 2 9 3 7 A 0 2 0 2 4

7. Question 6 · part b · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b only.

(c) **Figure 2** shows an algorithm that manipulates arrays.

The algorithm works with any number of scores.

```
1 SET oldScores TO [10, 20, 30, 40, 50]
2 SET newScores TO [0, 0, 0, 0, 0]
3 SET newIndex TO 0
4
5 FOR oldIndex FROM (LENGTH (oldScores) - 1) TO 0 STEP -1 DO
6     SET newScores[newIndex] TO oldScores[oldIndex]
7     SET newIndex TO newIndex + 1
8 END FOR
```

Figure 2

(i) Describe what happens to the variable **oldIndex** when line 5 is executed.

(2)

(ii) State the purpose of the algorithm in **Figure 2**.

(1)

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7. Question 6 · part b · 1 mark · Working and answer

Working and answer

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8. Question 6 · part c.i · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c.i only.

6 Programmers share algorithms with different people and write algorithms for different reasons.

(a) A programmer is showing a new algorithm to a group of non-technical managers.

State an appropriate method for writing the algorithm.

Justify your answer.

(2)

Method

Justification

(b) **Figure 1** shows an algorithm that displays a string based on the number input by the user.

```
1  SEND ("Enter a number: ") TO DISPLAY
2  RECEIVE inNum FROM (INTEGER) KEYBOARD
3  IF ((inNum = 1) OR (inNum = 2)) THEN
4      IF (inNum = 1) THEN
5          SEND ("First") TO DISPLAY
6      ELSE
7          IF (inNum = 2) THEN
8              SEND ("Second") TO DISPLAY
9          END IF
10     END IF
11 ELSE
12     SEND ("Invalid input") TO DISPLAY
13 END IF
```

Figure 1

Give **one** reason why the selection statement on line 7 is not required.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8. Question 6 · part c.i · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c.i only.

(c) **Figure 2** shows an algorithm that manipulates arrays.

The algorithm works with any number of scores.

```
1 SET oldScores TO [10, 20, 30, 40, 50]
2 SET newScores TO [0, 0, 0, 0, 0]
3 SET newIndex TO 0
4
5 FOR oldIndex FROM (LENGTH (oldScores) - 1) TO 0 STEP -1 DO
6     SET newScores[newIndex] TO oldScores[oldIndex]
7     SET newIndex TO newIndex + 1
8 END FOR
```

Figure 2

(i) Describe what happens to the variable **oldIndex** when line 5 is executed.

(2)

(ii) State the purpose of the algorithm in **Figure 2**.

(1)

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8. Question 6 · part c.i · 2 marks · Working and answer

Working and answer

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9. Question 6 · part c.ii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part c.ii only.

6 Programmers share algorithms with different people and write algorithms for different reasons.

(a) A programmer is showing a new algorithm to a group of non-technical managers.

State an appropriate method for writing the algorithm.

Justify your answer.

(2)

Method

Justification

(b) **Figure 1** shows an algorithm that displays a string based on the number input by the user.

```
1  SEND ("Enter a number: ") TO DISPLAY
2  RECEIVE inNum FROM (INTEGER) KEYBOARD
3  IF ((inNum = 1) OR (inNum = 2)) THEN
4      IF (inNum = 1) THEN
5          SEND ("First") TO DISPLAY
6      ELSE
7          IF (inNum = 2) THEN
8              SEND ("Second") TO DISPLAY
9          END IF
10     END IF
11 ELSE
12     SEND ("Invalid input") TO DISPLAY
13 END IF
```

Figure 1

Give **one** reason why the selection statement on line 7 is not required.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

9. Question 6 · part c.ii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part c.ii only.

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(c) **Figure 2** shows an algorithm that manipulates arrays.

The algorithm works with any number of scores.

```
1 SET oldScores TO [10, 20, 30, 40, 50]
2 SET newScores TO [0, 0, 0, 0, 0]
3 SET newIndex TO 0
4
5 FOR oldIndex FROM (LENGTH (oldScores) - 1) TO 0 STEP -1 DO
6     SET newScores[newIndex] TO oldScores[oldIndex]
7     SET newIndex TO newIndex + 1
8 END FOR
```

Figure 2

(i) Describe what happens to the variable **oldIndex** when line 5 is executed.

(2)

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(ii) State the purpose of the algorithm in **Figure 2**.

(1)

P 7 2 9 3 7 A 0 2 1 2 4

21

Turn over ▶

9. Question 6 · part c.ii · 1 mark

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Further practice · Attempt part c.ii only.

(d) Cloud storage has become an increasingly popular service.

Discuss the benefits and drawbacks of using cloud storage.

(6)

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(Total for Question 6 = 12 marks)

9. Question 6 · part c.ii · 1 mark · Working and answer

Working and answer

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10. Question 6 · part c · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c only.

6 Pseudocode and flowcharts are used to create algorithms.

(a) The number of minutes a train is early or late are recorded each day for a week.

- A value of 0 is recorded if the train is on time.
- A negative value is recorded if the train is early.
- A positive value is recorded if the train is late.

The algorithm in **Figure 2** has been created to calculate and output the total number of trains that were early, on time or late.

```
01
02  SET arrival TO [-2, 1, 5, 0, -3, 4, 1]
03
04  SET early TO 0
05  SET late TO 0
06  SET index TO 0
07
08  WHILE NOT (index > LENGTH(arrival)) DO
09
10      IF arrival[index] >= 0 THEN
11          SET late TO arrival[index]
12      ELSE
13          IF arrival[index] < 0 THEN
14              SET early TO early + 1
15          END IF
16      END IF
17
18      SET index TO index + 1
19
20  END WHILE
21
22  SEND "Trains early:    " & early TO DISPLAY
23  SEND "Trains on time: " & (late + early) TO DISPLAY
24  SEND "Trains late:     " & late TO DISPLAY
25
```

Figure 2

Identify the type of iteration construct used in **Figure 2**.

(1)

- ☒ **A** Post-conditioned loop
- ☒ **B** Count controlled loop
- ☒ **C** Array loop
- ☒ **D** Pre-conditioned loop

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10. Question 6 · part c · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c only.

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(b) There are **four** errors in the pseudocode in **Figure 2**.

Write the correct code for lines 08, 10, 11 and 23

(i) Line 08

(1)

(ii) Line 10

(1)

(iii) Line 11

(1)

(iv) Line 23

(1)

(c) Give **one** reason for the variable `index` being incremented in line 18.

(1)

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10. Question 6 · part c · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c only.

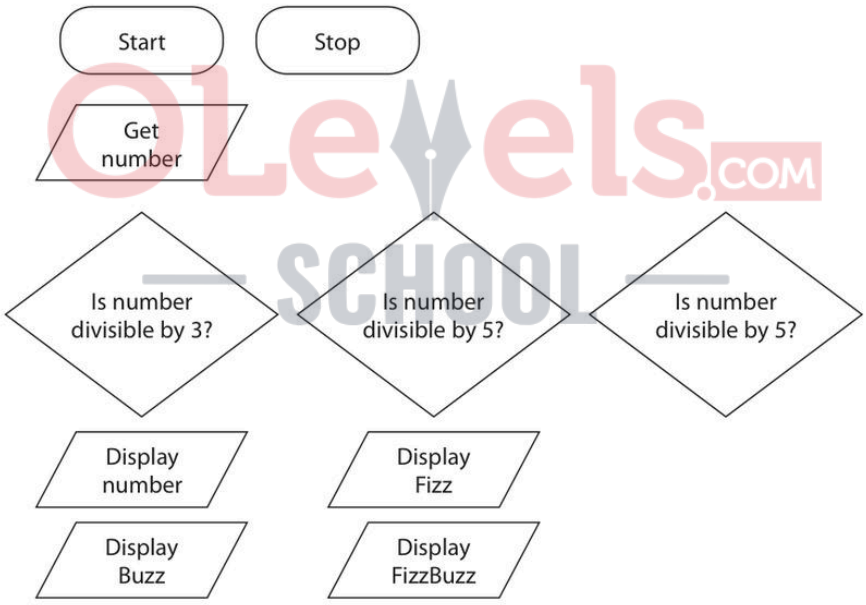
(d) In the game FizzBuzz a user enters a number.

These conditions apply:

- if the number is divisible by 3 the program displays Fizz
- if the number is divisible by 5 the program displays Buzz
- if the number is divisible by both 3 and 5 the program displays FizzBuzz
- if the number is not divisible by either 3 or 5, the program displays the number entered.

Hint: If one number is divisible by another number there is no remainder.

These are the components needed for the flowchart.



Draw the flowchart for the algorithm used in the game in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

(6)

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P 7 5 7 3 6 A 0 2 4 2 8

10. Question 6 · part c · 1 mark · Working and answer

Working and answer

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11. Question 6 · part b.iii · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.iii only.

6 Algorithms can be represented using flowcharts or pseudocode.

(a) An algorithm has been designed to perform these steps:

- A user is asked to input a number **n**
- If the number **n** is 0 the program stops
- If the number **n** is positive a message to display 'Positive' is output
- If the number **n** is negative a message to display 'Negative' is output
- The program repeats until a value of 0 has been input.

Draw a flowchart to represent this algorithm.

(6)

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11. Question 6 · part b.iii · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.iii only.

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(b) **Figure 3** shows an algorithm written in pseudocode that performs a mathematical process.

```
01
02  RECEIVE number FROM (INTEGER) KEYBOARD
03
04  SET flag TO FALSE
05
06  IF number = 1 THEN
07     SET flag TO TRUE
08  ELSE
09     SET testNum TO 1
10     REPEAT
11        SET testNum TO testNum + 1
12        IF (number MOD testNum) = 0 THEN
13           SET flag TO TRUE
14        END IF
15     UNTIL (testNum = number) OR (flag = TRUE)
16  END IF
17
18  IF NOT (flag = TRUE)
19     SEND number TO DISPLAY
20  END IF
21
```

Figure 3

Trace tables are used to show the outputs from algorithms.

(i) Complete the trace table for an input of 7

(2)

number	flag	testNum	DISPLAY

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11. Question 6 · part b.iii · 1 mark

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Further practice · Attempt part b.iii only.

(ii) Complete the trace table for an input of 9

(2)

number	flag	testNum	DISPLAY

(iii) State the purpose of the algorithm.

(1)

.....

.....

(iv) Lines 10 to 15 of the pseudocode in **Figure 3** use a post-conditioned loop.

Explain **one** reason why it is better for the algorithm to use a post-conditioned loop rather than a count controlled loop.

(2)

.....

.....

.....

.....

.....

(Total for Question 6 = 13 marks)

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11. Question 6 · part b.iii · 1 mark · Working and answer

Working and answer

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Practice workbook · Find and fix errors in an algorithm

Logic and off-by-one errors in supplied pseudocode or flowchart algorithms: locating the faulty line, explaining the wrong result it causes, and writing a corrected line.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-04

1. Question 6 · part b.iii · 1 mark

2024 JUN · Paper 01

Worked-example retry · Attempt part b.iii only.

6 Pseudocode and flowcharts are used to create algorithms.

(a) The number of minutes a train is early or late are recorded each day for a week.

- A value of 0 is recorded if the train is on time.
- A negative value is recorded if the train is early.
- A positive value is recorded if the train is late.

The algorithm in **Figure 2** has been created to calculate and output the total number of trains that were early, on time or late.

```
01
02  SET arrival TO [-2, 1, 5, 0, -3, 4, 1]
03
04  SET early TO 0
05  SET late TO 0
06  SET index TO 0
07
08  WHILE NOT (index > LENGTH(arrival)) DO
09
10      IF arrival[index] >= 0 THEN
11          SET late TO arrival[index]
12      ELSE
13          IF arrival[index] < 0 THEN
14              SET early TO early + 1
15          END IF
16      END IF
17
18      SET index TO index + 1
19
20  END WHILE
21
22  SEND "Trains early:    " & early TO DISPLAY
23  SEND "Trains on time: " & (late + early) TO DISPLAY
24  SEND "Trains late:     " & late TO DISPLAY
25
```

Figure 2

Identify the type of iteration construct used in **Figure 2**.

(1)

- ☒ **A** Post-conditioned loop
- ☒ **B** Count controlled loop
- ☒ **C** Array loop
- ☒ **D** Pre-conditioned loop

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 5 7 3 6 A 0 2 2 2 8

1. Question 6 · part b.iii · 1 mark

2024 JUN · Paper 01

Worked-example retry · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) There are **four** errors in the pseudocode in **Figure 2**.

Write the correct code for lines 08, 10, 11 and 23

(i) Line 08

(1)

(ii) Line 10

(1)

(iii) Line 11

(1)

(iv) Line 23

(1)

(c) Give **one** reason for the variable `index` being incremented in line 18.

(1)

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1. Question 6 · part b.iii · 1 mark · Working and answer

Working and answer

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2. Question 5 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

5 Ships carry cargo around the world in containers.

(a) Containers come in two sizes.

Figure 2 shows an algorithm written using flowchart symbols.

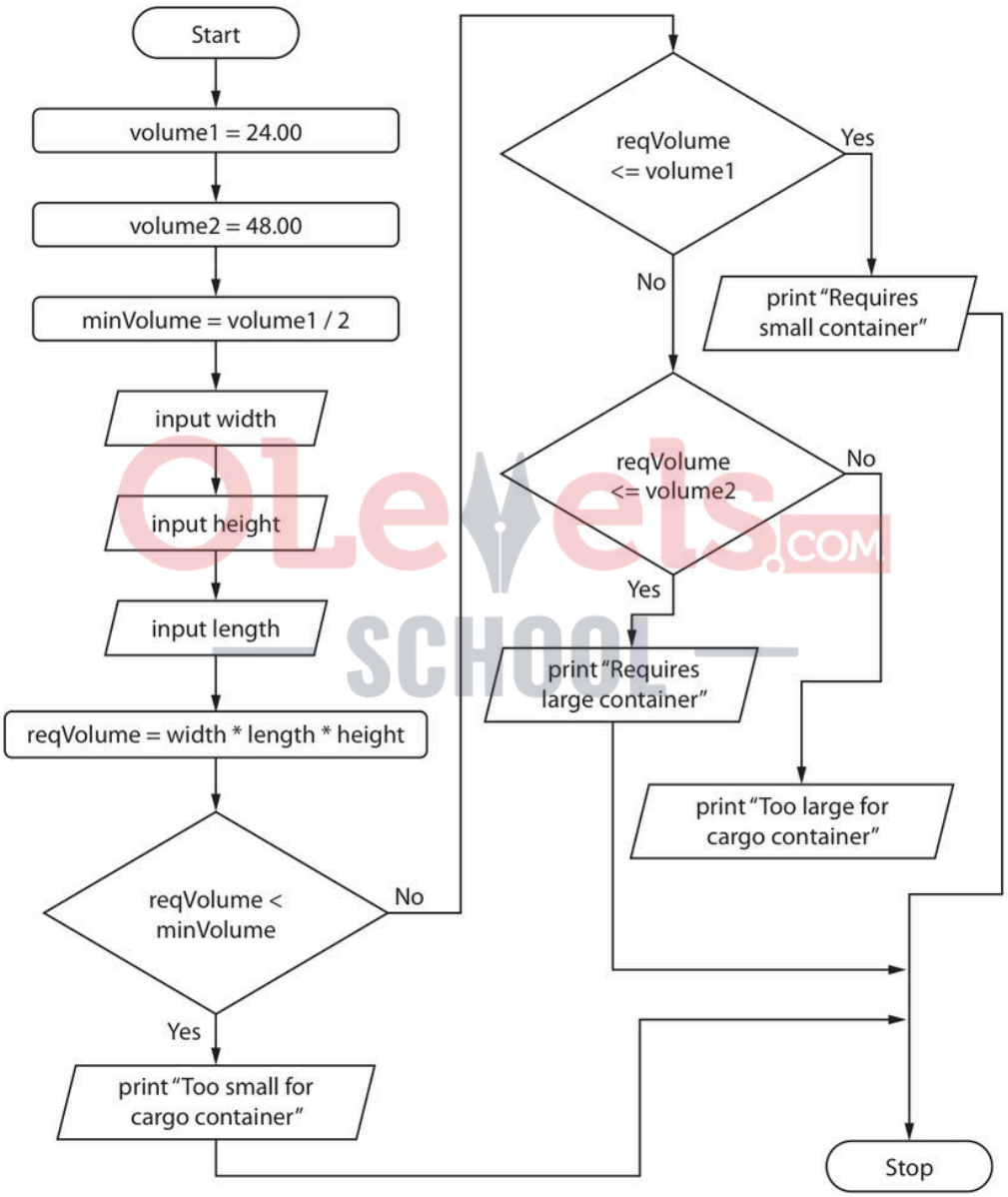


Figure 2

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DO NOT WRITE IN THIS AREA

2. Question 5 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

(i) Complete the table to show the output for each cargo item.

(3)

Cargo item			Output
width	length	height	
4	4	2	
2	2	2	
3	8	5	

(ii) State the purpose of the algorithm in **Figure 2**.

(1)

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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

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2. Question 5 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

(b) Each ship is registered to a state.

Figure 3 shows a partially completed algorithm written in pseudocode.

The completed algorithm must:

- print each state to the display on a new line
- count the number of states
- create a message as a single string (e.g. there are *number* states)
- print the message to the display.

Complete the algorithm in the space provided in **Figure 3**.

(4)

```
SET numStates TO 0
SET states TO ["France", "Singapore", "Malta", "Panama", "Greece", "Italy"]

FOR EACH state FROM states DO




END FOREACH
```

Figure 3

DO NOT WRITE IN THIS AREA
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DO NOT WRITE IN THIS AREA

2. Question 5 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

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2. Question 5 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

(c) Cargo ships have maximum weight loads.

Figure 4 shows an algorithm written using pseudocode.

The algorithm should identify the size of cargo ship required for any load.

There is an error on line 9.

```
2   SET loadWeight TO [20000, 28000, 40000, 50000]
3   SET index TO 0
4   SET found TO FALSE
5
6   SEND "Enter cargo weight" TO DISPLAY
7   RECEIVE target FROM (INTEGER) KEYBOARD
8
9   WHILE (NOT found) DO
10      IF (loadWeight [index] >= target) THEN
11          SEND loadWeight [index] TO DISPLAY
12          SET found TO TRUE
13      ELSE
14          SET index TO  index + 1
15      END IF
16  END WHILE
17
18  IF (NOT found) THEN
19      SEND "No ship available" TO DISPLAY
20  END IF
```

Figure 4

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

2. Question 5 · part c.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.ii only.

(i) Trace tables are used to identify errors in algorithms.

Complete the trace table for an input of 50500 to show what happens due to the error on line 9 in the pseudocode in **Figure 4**.

You may not need to fill in all the rows in the table.

(2)

target	found	index	loadWeight[index]
50500	FALSE	0	

(ii) Construct a single line of pseudocode to correct line 9.

(2)

.....

.....

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.....

(Total for Question 5 = 12 marks)

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2. Question 5 · part c.ii · 2 marks · Working and answer

Working and answer

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3. Question 5 · part b.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b.i only.

5 Isaac is a program developer.

(a) **Figure 3** shows an algorithm Isaac has written.

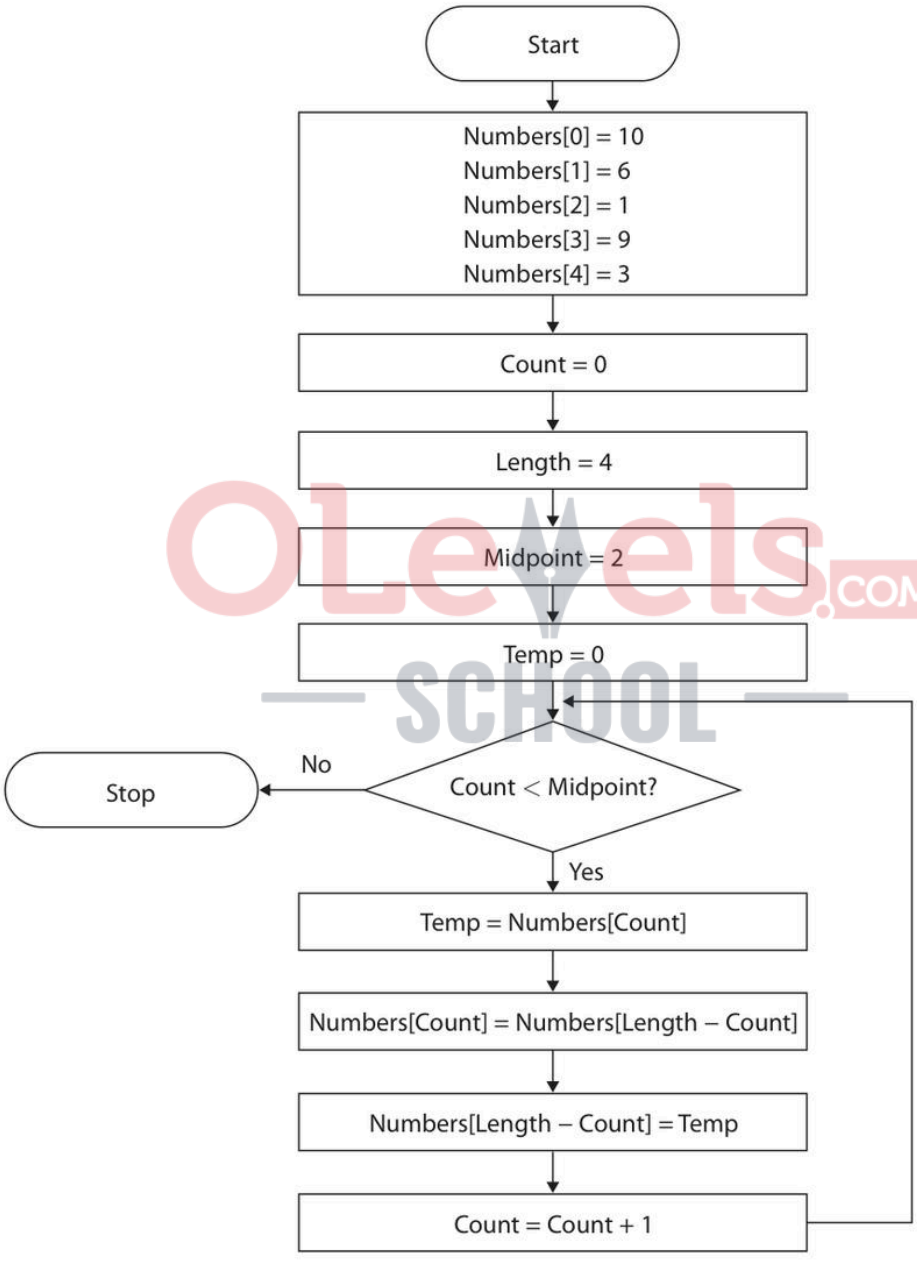


Figure 3

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 6 4 8 9 A 0 1 2 2 0

3. Question 5 · part b.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(i) Complete the trace table. You may not need to use all of the rows.

(5)

				Numbers array				
Count	**Length**	**Midpoint**	**Temp**	[0]	[1]	[2]	[3]	[4]
0	4	2	0	10	6	1	9	3

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(ii) Give the purpose of the algorithm.

(1)

(iii) Explain why the variable Temp is needed.

(2)

3. Question 5 · part b.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b.i only.

(b) **Figure 4** shows an algorithm Isaac has written using pseudocode.

The algorithm should display the average of the numbers that have been input.

```
1   SET total TO 0
2   SET number TO 0
3   SET count TO 0
4   WHILE number <> –1 DO
5      SEND 'Input a number or –1 to end the program' TO DISPLAY
6      RECEIVE number FROM (INTEGER) KEYBOARD
7      SET total TO total + number
8      SET count TO count + 1
9   END WHILE
10  SET average TO total / count
11  SEND 'The average is ' & average TO DISPLAY
```

Figure 4

Isaac uses the input 2, 3, 5, 2, –1 to test the algorithm. He discovers an error.

Expected result	Actual result
The average is 2.75	The average is 2.2

(i) Explain why the **Actual** result is not the same as the **Expected** result.

(2)

(ii) Give the number of the line that contains the error.

(1)

(iii) Amend a single line of pseudocode to correct the error.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 6 6 4 8 9 A 0 1 4 2 0

3. Question 5 · part b.i · 2 marks · Working and answer

Working and answer

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4. Question 5 · part b.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b.ii only.

5 Isaac is a program developer.

(a) **Figure 3** shows an algorithm Isaac has written.

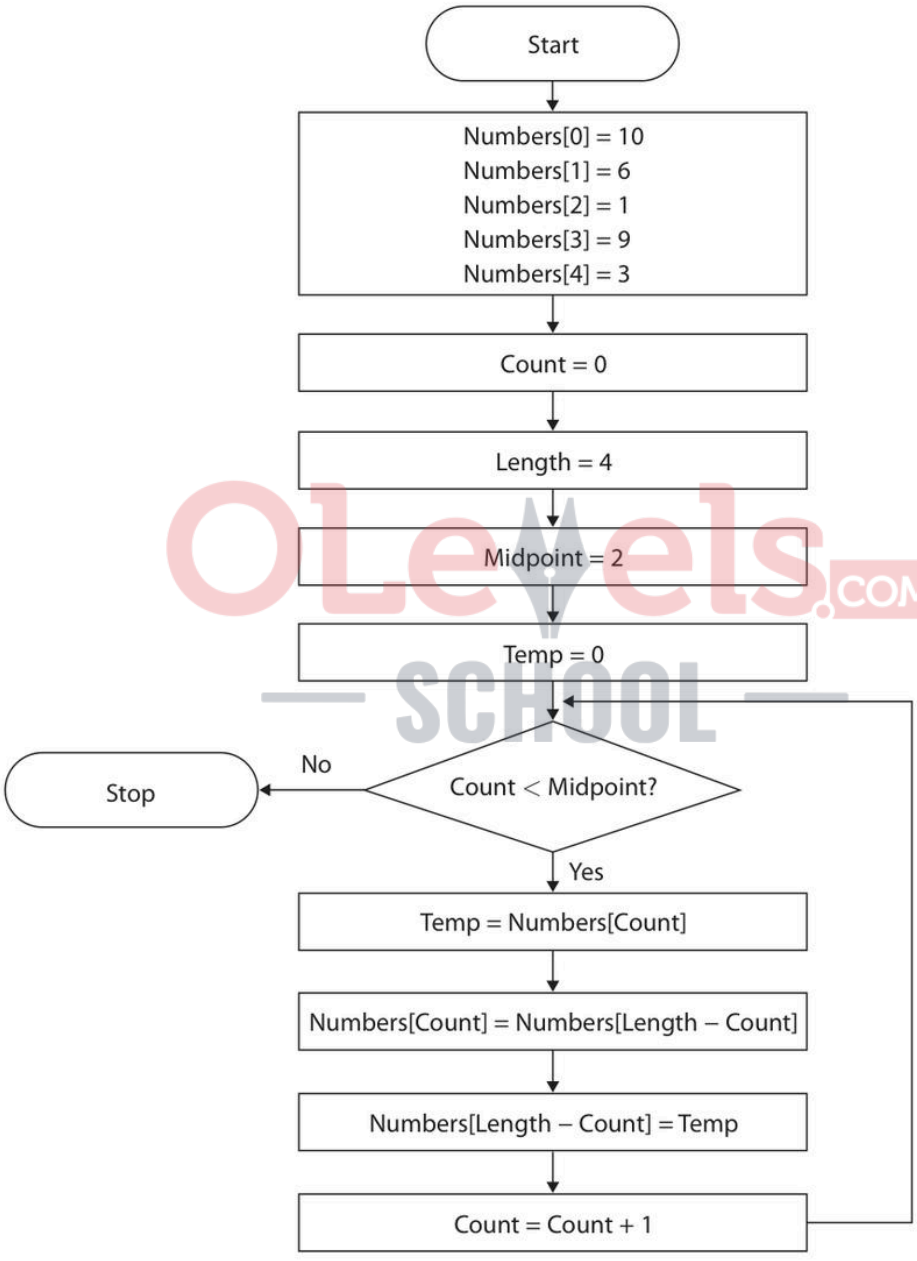


Figure 3

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DO NOT WRITE IN THIS AREA

P 6 6 4 8 9 A 0 1 2 2 0

4. Question 5 · part b.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(i) Complete the trace table. You may not need to use all of the rows.

(5)

				Numbers array				
Count	**Length**	**Midpoint**	**Temp**	[0]	[1]	[2]	[3]	[4]
0	4	2	0	10	6	1	9	3

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(ii) Give the purpose of the algorithm.

(1)

(iii) Explain why the variable Temp is needed.

(2)

4. Question 5 · part b.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b.ii only.

(b) **Figure 4** shows an algorithm Isaac has written using pseudocode.

The algorithm should display the average of the numbers that have been input.

```
1   SET total TO 0
2   SET number TO 0
3   SET count TO 0
4   WHILE number <> –1 DO
5      SEND 'Input a number or –1 to end the program' TO DISPLAY
6      RECEIVE number FROM (INTEGER) KEYBOARD
7      SET total TO total + number
8      SET count TO count + 1
9   END WHILE
10  SET average TO total / count
11  SEND 'The average is ' & average TO DISPLAY
```

Figure 4

Isaac uses the input 2, 3, 5, 2, –1 to test the algorithm. He discovers an error.

Expected result	Actual result
The average is 2.75	The average is 2.2

(i) Explain why the **Actual** result is not the same as the **Expected** result.

(2)

(ii) Give the number of the line that contains the error.

(1)

(iii) Amend a single line of pseudocode to correct the error.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

14

P 6 6 4 8 9 A 0 1 4 2 0

4. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

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5. Question 5 · part b.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b.iii only.

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(a) **Figure 3** shows an algorithm Isaac has written.

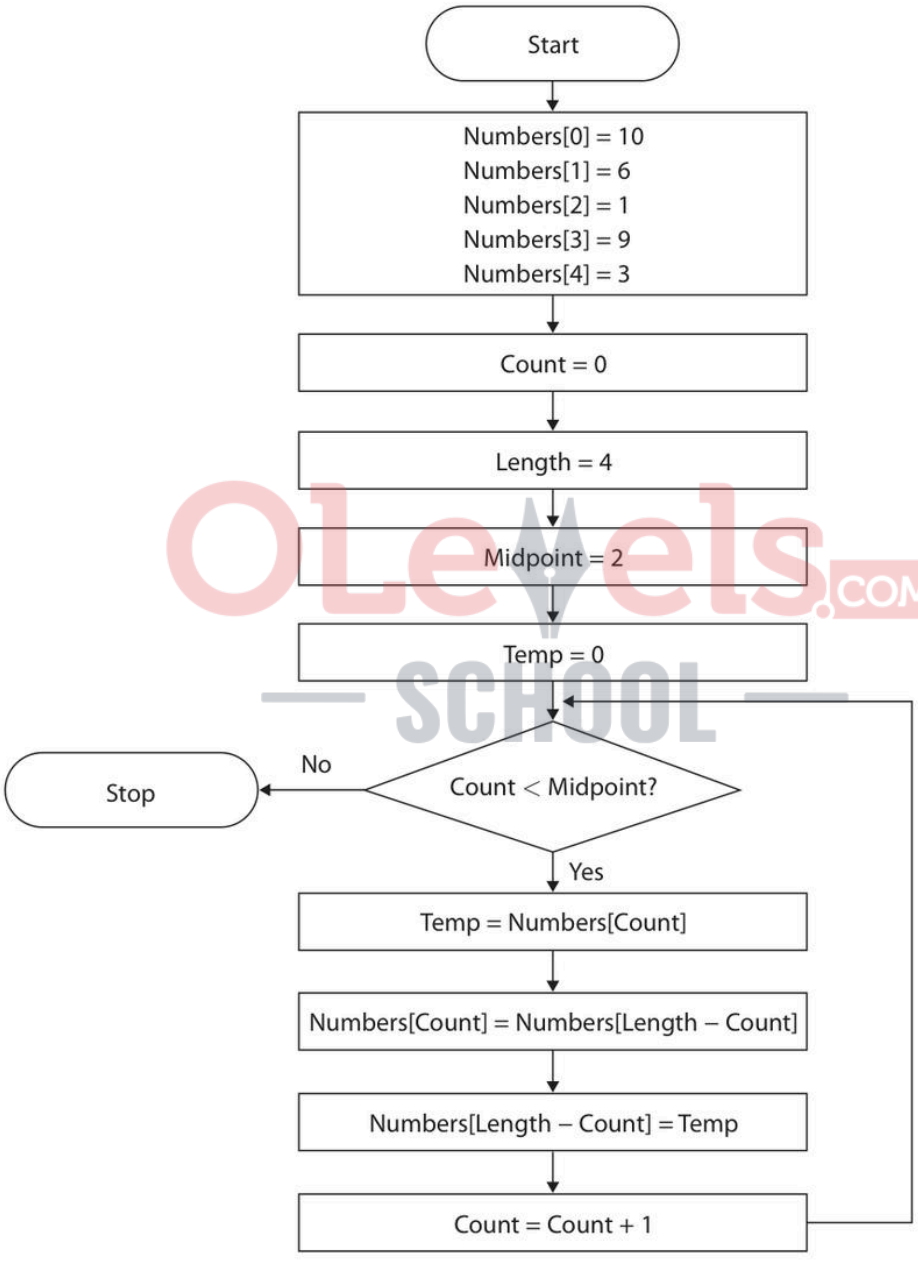


Figure 3

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DO NOT WRITE IN THIS AREA

5. Question 5 · part b.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(i) Complete the trace table. You may not need to use all of the rows.

(5)

				Numbers array				
Count	**Length**	**Midpoint**	**Temp**	[0]	[1]	[2]	[3]	[4]
0	4	2	0	10	6	1	9	3

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(ii) Give the purpose of the algorithm.

(1)

(iii) Explain why the variable Temp is needed.

(2)

5. Question 5 · part b.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b.iii only.

(b) **Figure 4** shows an algorithm Isaac has written using pseudocode.

The algorithm should display the average of the numbers that have been input.

```
1   SET total TO 0
2   SET number TO 0
3   SET count TO 0
4   WHILE number <> –1 DO
5      SEND 'Input a number or –1 to end the program' TO DISPLAY
6      RECEIVE number FROM (INTEGER) KEYBOARD
7      SET total TO total + number
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```

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Isaac uses the input 2, 3, 5, 2, –1 to test the algorithm. He discovers an error.

Expected result	Actual result
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(2)

(ii) Give the number of the line that contains the error.

(1)

(iii) Amend a single line of pseudocode to correct the error.

(1)

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DO NOT WRITE IN THIS AREA

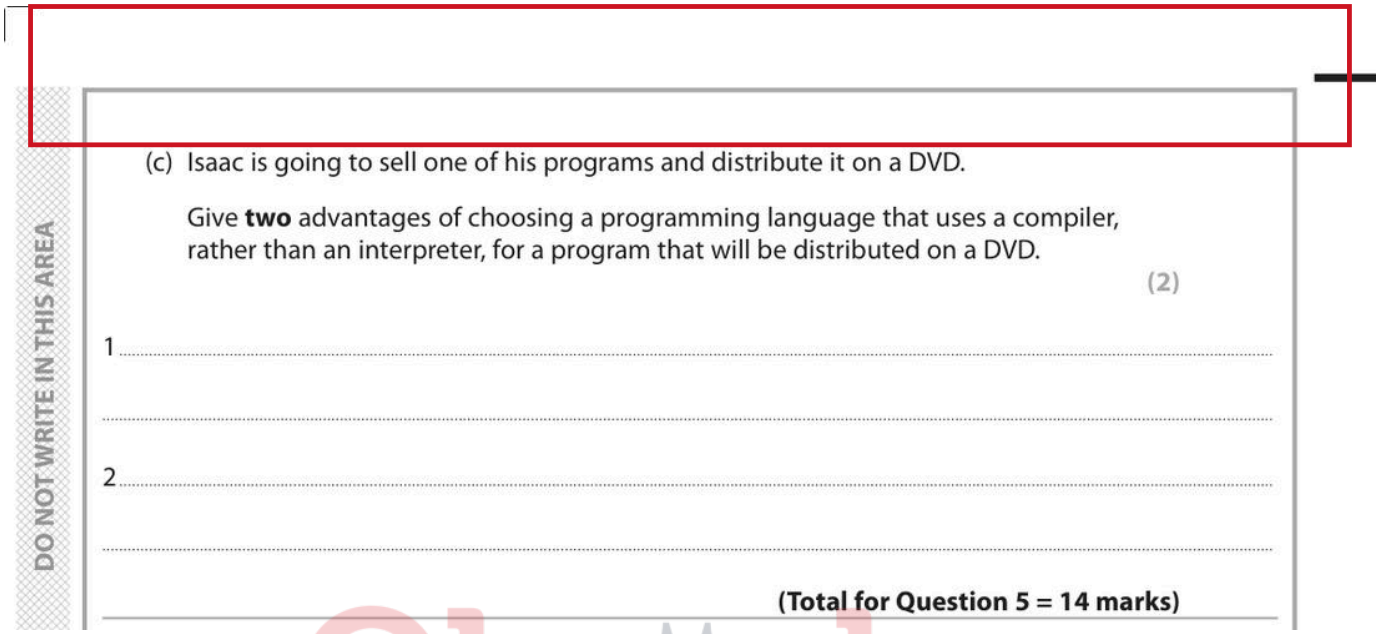
14

P 6 6 4 8 9 A 0 1 4 2 0

5. Question 5 · part b.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b.iii only.



DO NOT WRITE IN THIS AREA

(c) Isaac is going to sell one of his programs and distribute it on a DVD.

Give **two** advantages of choosing a programming language that uses a compiler, rather than an interpreter, for a program that will be distributed on a DVD.

(2)

1 .....

.....

2 .....

.....

(Total for Question 5 = 14 marks)

Working and answer

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6. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 01

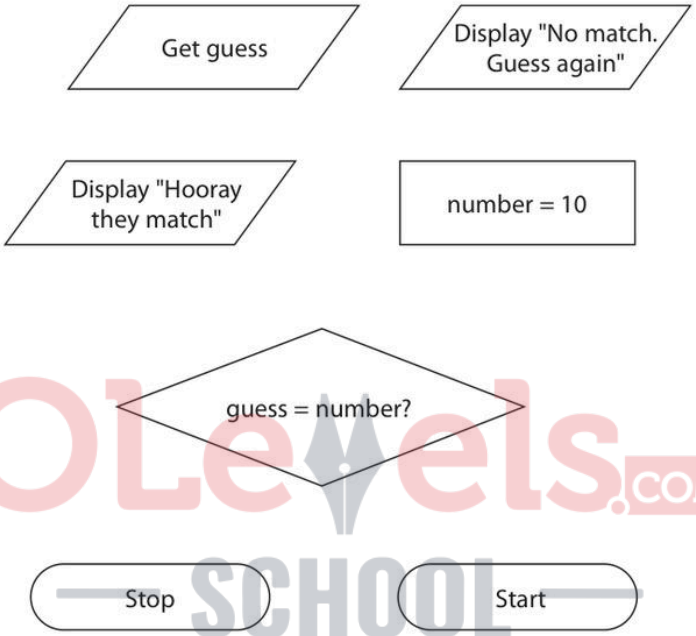
Further practice · Attempt part c.ii only.

4 Reba likes writing programs.

(a) She is writing a guessing game.

She needs a flowchart to show the logic of the game.

(i) These are the components needed to draw the flowchart.



Draw the flowchart for the algorithm in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

(5)

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DO NOT WRITE IN THIS AREA

6. Question 4 · part c.ii · 1 mark

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Further practice · Attempt part c.ii only.

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Draw your flowchart here.

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6. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

(ii) Identify an alternative method for writing the algorithm.

(1)

- ☒ **A** Simulation
- ☒ **B** Cipher
- ☒ **C** Program code
- ☒ **D** Truth table

(b) Reba wants to develop a program that will convert a temperature in Fahrenheit to Celsius.

Here are four steps in the algorithm.

The steps are not in the correct order.

	Step
A	Change the temperature to Celsius
B	Get the temperature in Fahrenheit
C	Show the temperature in Celsius
D	Set the temperature to 0

(i) Give the letter of the step that initialises a variable.

(1)

(ii) Give the letter of the step that inputs a value.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 2 5 3 8 A 0 1 2 2 0

6. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

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QUESTION 4 (c) BEGINS ON THE NEXT PAGE.

6. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

(c) **Figure 2** shows the pseudocode for an early version of an algorithm that Reba has written for another game.

The algorithm:

- asks the user to input a colour or input –1 to end the game
- awards 1 point for red
- awards 8 points for orange
- generates the score for the game
- displays the results of the game.

```
SET Colour TO ""
SET Score TO 0
SET RedPoints TO 0
SET OrangePoints TO 0
SET NumOranges TO 0

WHILE Colour <> "-1" DO
   RECEIVE Colour FROM (STRING) KEYBOARD
   IF Colour = "red" THEN
      SET RedPoints TO RedPoints + 1
   ELSE
      IF Colour = "orange" THEN
         SET OrangePoints TO OrangePoints + 8
         SET NumOranges TO NumOranges + 1
      END IF
   END IF
END WHILE

SET Score TO RedPoints + OrangePoints

SEND ("Score: "& Score) TO DISPLAY
SEND ("Number of reds: "& RedPoints) TO DISPLAY
SEND ("Number of oranges: "& OrangePoints) TO DISPLAY
```

Figure 2

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 2 5 3 8 A 0 1 4 2 0

6. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

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Reba inputs: red, orange, red, red, orange, –1

The outputs are not as she expects.

(i) Complete the trace table to show the outputs.

(4)

Colour	Score	RedPoints	OrangePoints	NumOranges	Outputs
	0	0	0	0	
red					
orange					
red					
red					
orange					
–1					

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(ii) Give the line number of the pseudocode that contains the error.

(1)

.....

(iii) Write a replacement line of pseudocode to correct the error.

(1)

.....

(Total for Question 4 = 14 marks)

6. Question 4 · part c.ii · 1 mark · Working and answer

Working and answer

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7. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 01

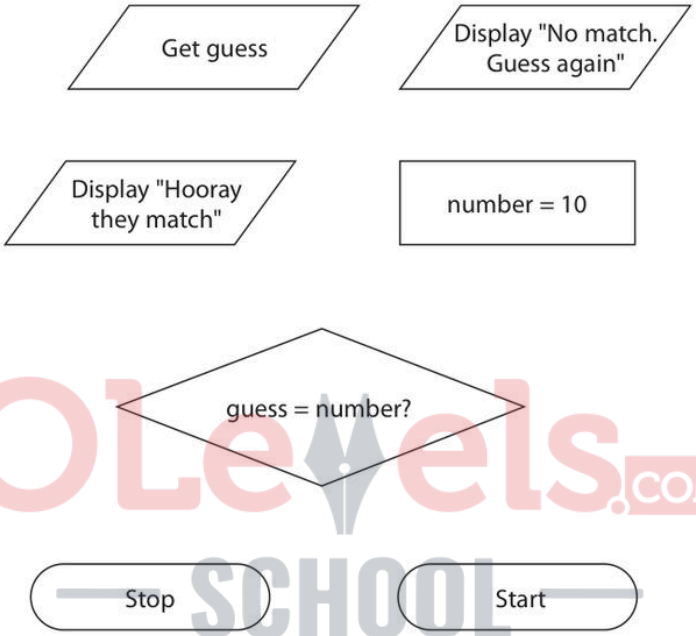
Further practice · Attempt part c.iii only.

4 Reba likes writing programs.

(a) She is writing a guessing game.

She needs a flowchart to show the logic of the game.

(i) These are the components needed to draw the flowchart.



Draw the flowchart for the algorithm in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

(5)

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P 7 2 5 3 8 A 0 1 0 2 0

7. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.iii only.

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DO NOT WRITE IN THIS AREA

Draw your flowchart here.

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7. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.iii only.

(ii) Identify an alternative method for writing the algorithm.

(1)

- ☒ **A** Simulation
- ☒ **B** Cipher
- ☒ **C** Program code
- ☒ **D** Truth table

(b) Reba wants to develop a program that will convert a temperature in Fahrenheit to Celsius.

Here are four steps in the algorithm.

The steps are not in the correct order.

	Step
A	Change the temperature to Celsius
B	Get the temperature in Fahrenheit
C	Show the temperature in Celsius
D	Set the temperature to 0

(i) Give the letter of the step that initialises a variable.

(1)

(ii) Give the letter of the step that inputs a value.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 2 5 3 8 A 0 1 2 2 0

7. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.iii only.

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QUESTION 4 (c) BEGINS ON THE NEXT PAGE.

7. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.iii only.

(c) **Figure 2** shows the pseudocode for an early version of an algorithm that Reba has written for another game.

The algorithm:

- asks the user to input a colour or input –1 to end the game
- awards 1 point for red
- awards 8 points for orange
- generates the score for the game
- displays the results of the game.

```
SET Colour TO ""
SET Score TO 0
SET RedPoints TO 0
SET OrangePoints TO 0
SET NumOranges TO 0

WHILE Colour <> "-1" DO
   RECEIVE Colour FROM (STRING) KEYBOARD
   IF Colour = "red" THEN
      SET RedPoints TO RedPoints + 1
   ELSE
      IF Colour = "orange" THEN
         SET OrangePoints TO OrangePoints + 8
         SET NumOranges TO NumOranges + 1
      END IF
   END IF
END WHILE

SET Score TO RedPoints + OrangePoints

SEND ("Score: "& Score) TO DISPLAY
SEND ("Number of reds: "& RedPoints) TO DISPLAY
SEND ("Number of oranges: "& OrangePoints) TO DISPLAY
```

Figure 2

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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 2 5 3 8 A 0 1 4 2 0

7. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.iii only.

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DO NOT WRITE IN THIS AREA

Reba inputs: red, orange, red, red, orange, –1

The outputs are not as she expects.

(i) Complete the trace table to show the outputs.

(4)

Colour	Score	RedPoints	OrangePoints	NumOranges	Outputs
	0	0	0	0	
red					
orange					
red					
red					
orange					
–1					

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(ii) Give the line number of the pseudocode that contains the error.

(1)

.....

(iii) Write a replacement line of pseudocode to correct the error.

(1)

.....

(Total for Question 4 = 14 marks)

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7. Question 4 · part c.iii · 1 mark · Working and answer

Working and answer

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8. Question 5 · part b · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

5 A marine scientist is conducting a study of tuna in the world's oceans.

(a) A list of tuna species needs to be sorted into **descending** alphabetical order using a merge sort algorithm.

Figure 5 shows the lists created at the end of the splitting process.

Each list is a single tuna species.

Bigeye	Blackfin	Albacore	Longtail	Bluefin

Figure 5

It will require three passes to merge the lists into a single sorted list in **descending** order.

Complete the merge sort using the space provided.

(2)

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8. Question 5 · part b · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

(b) A bubble sort could be used to sort the list of tuna species into **ascending** order.

Figure 6 shows an algorithm for a bubble sort.

```
1  SET myTuna TO ["Bigeye", "Blackfin", "Albacore",
2                 "Longtail", "Bluefin"]
3  SET tmp TO 0
4  SET swaps TO True
5  SET length TO LENGTH (myTuna)
6
7  WHILE (swaps = True) DO
8      SET swaps TO False
9      FOR ndx FROM 0 TO length - 1 DO
10         IF myTuna[ndx] > myTuna[ndx + 1] THEN
11             SET tmp TO myTuna[ndx + 1]
12             SET myTuna[ndx] TO myTuna[ndx + 1]
13             SET myTuna[ndx + 1] TO tmp
14             SET swaps TO True
15         END IF
16     END FOR
17 END WHILE
```

Figure 6

There is an error in the loop between line 9 and line 16.

Complete the table to give the line number with an error and a corrected line of pseudocode.

(2)

Line number with error	
Corrected line of pseudocode	

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

12

P 7 2 9 3 8 A 0 1 2 1 6

8. Question 5 · part b · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

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(c) The scientist is collecting and storing data about tuna.

The data collected is:

- species
- length in centimetres
- weight in kilograms
- age in years.

The data is stored in an array.

The collected data needs to be written to a file named **TunaData.txt**

Each record stored in the file must have a code number in the first field.

Code numbers must start at 101

Each field in a record should be separated by a comma.

Figure 7 shows the contents of the file.

```
101,Yellowfin,105,15,3
102,Albacore,90,15,5
103,Skipjack,50,3,4
104,Bigeye,105,25,4
105,Atlantic Bonito,50,4,2
106,Northern Bluefin,190,120,11
107,Southern Bluefin,190,120,11
108,Tongol,90,20,4
```

Figure 7

Open **Q05c** in the code editor.

Write the program.

You must use the structure given in **Q05c** to write the program.

Do not add any further functionality.

Save your code as **Q05cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 5 = 10 marks)

8. Question 5 · part b · 2 marks · Working and answer

Working and answer

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9. Question 6 · part b.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b.i only.

6 Pseudocode and flowcharts are used to create algorithms.

(a) The number of minutes a train is early or late are recorded each day for a week.

- A value of 0 is recorded if the train is on time.
- A negative value is recorded if the train is early.
- A positive value is recorded if the train is late.

The algorithm in **Figure 2** has been created to calculate and output the total number of trains that were early, on time or late.

```
01
02  SET arrival TO [-2, 1, 5, 0, -3, 4, 1]
03
04  SET early TO 0
05  SET late TO 0
06  SET index TO 0
07
08  WHILE NOT (index > LENGTH(arrival)) DO
09
10      IF arrival[index] >= 0 THEN
11          SET late TO arrival[index]
12      ELSE
13          IF arrival[index] < 0 THEN
14              SET early TO early + 1
15          END IF
16      END IF
17
18      SET index TO index + 1
19
20  END WHILE
21
22  SEND "Trains early:    " & early TO DISPLAY
23  SEND "Trains on time: " & (late + early) TO DISPLAY
24  SEND "Trains late:     " & late TO DISPLAY
25
```

Figure 2

Identify the type of iteration construct used in **Figure 2**.

(1)

- ☒ **A** Post-conditioned loop
- ☒ **B** Count controlled loop
- ☒ **C** Array loop
- ☒ **D** Pre-conditioned loop

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9. Question 6 · part b.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b.i only.

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) There are **four** errors in the pseudocode in **Figure 2**.

Write the correct code for lines 08, 10, 11 and 23

(i) Line 08

(1)

(ii) Line 10

(1)

(iii) Line 11

(1)

(iv) Line 23

(1)

(c) Give **one** reason for the variable `index` being incremented in line 18.

(1)

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9. Question 6 · part b.i · 1 mark · Working and answer

Working and answer

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10. Question 6 · part b.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b.ii only.

6 Pseudocode and flowcharts are used to create algorithms.

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05  SET late TO 0
06  SET index TO 0
07
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09
10      IF arrival[index] >= 0 THEN
11          SET late TO arrival[index]
12      ELSE
13          IF arrival[index] < 0 THEN
14              SET early TO early + 1
15          END IF
16      END IF
17
18      SET index TO index + 1
19
20  END WHILE
21
22  SEND "Trains early:    " & early TO DISPLAY
23  SEND "Trains on time: " & (late + early) TO DISPLAY
24  SEND "Trains late:     " & late TO DISPLAY
25
```

Figure 2

Identify the type of iteration construct used in **Figure 2**.

(1)

- ☒ **A** Post-conditioned loop
- ☒ **B** Count controlled loop
- ☒ **C** Array loop
- ☒ **D** Pre-conditioned loop

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 5 7 3 6 A 0 2 2 2 8

10. Question 6 · part b.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) There are **four** errors in the pseudocode in **Figure 2**.

Write the correct code for lines 08, 10, 11 and 23

(i) Line 08

(1)

(ii) Line 10

(1)

(iii) Line 11

(1)

(iv) Line 23

(1)

(c) Give **one** reason for the variable `index` being incremented in line 18.

(1)

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10. Question 6 · part b.ii · 1 mark · Working and answer

Working and answer

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11. Question 6 · part b.iv · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b.iv only.

6 Pseudocode and flowcharts are used to create algorithms.

(a) The number of minutes a train is early or late are recorded each day for a week.

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08  WHILE NOT (index > LENGTH(arrival)) DO
09
10      IF arrival[index] >= 0 THEN
11          SET late TO arrival[index]
12      ELSE
13          IF arrival[index] < 0 THEN
14              SET early TO early + 1
15          END IF
16      END IF
17
18      SET index TO index + 1
19
20  END WHILE
21
22  SEND "Trains early:    " & early TO DISPLAY
23  SEND "Trains on time: " & (late + early) TO DISPLAY
24  SEND "Trains late:     " & late TO DISPLAY
25
```

Figure 2

Identify the type of iteration construct used in **Figure 2**.

(1)

- ☒ **A** Post-conditioned loop
- ☒ **B** Count controlled loop
- ☒ **C** Array loop
- ☒ **D** Pre-conditioned loop

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11. Question 6 · part b.iv · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b.iv only.

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(b) There are **four** errors in the pseudocode in **Figure 2**.

Write the correct code for lines 08, 10, 11 and 23

(i) Line 08

(1)

(ii) Line 10

(1)

(iii) Line 11

(1)

(iv) Line 23

(1)

(c) Give **one** reason for the variable `index` being incremented in line 18.

(1)

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11. Question 6 · part b.iv · 1 mark · Working and answer

Working and answer

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12. Question 5 · part a.i · 2 marks

2025 JUN · Paper 02

Further practice · Attempt part a.i only.

5 Martha is a computer science student and is learning to code.

(a) **Figure 7** shows an algorithm in pseudocode written by Martha.

```
1  SET first TO 0
2  SET second TO 0
3  SET answer TO 0
4  SET user_answer TO 0
5  SET score TO 0
6
7  REPEAT 10 TIMES
8      SET score TO 0
9      SET first TO RANDOM(99)
10     SET second TO RANDOM(99)
11     SET answer TO first + second
12
13     SEND first "+" second "=" TO DISPLAY
14     RECEIVE user_answer FROM (INTEGER) KEYBOARD
15
16     IF user_answer = answer THEN
17        SET score TO score + 1
18     END IF
19 END REPEAT
20 SEND score TO DISPLAY
```

Figure 7

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12. Question 5 · part a.i · 2 marks

2025 JUN · Paper 02

Further practice · Attempt part a.i only.

There is a logic error on line 8 of the pseudocode.

(i) Explain the problem this error will cause.

(2)

(ii) Describe the purpose of the algorithm shown in **Figure 7**, assuming the error has been corrected.

(2)

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12. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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